What Four Research Attempts Taught Us

Plain-language main talk with technical appendix

2026-07-18

1. What We Learned From Four Research Attempts
2. Technical Appendix
3. Post-SAQ Pivot: Attempts 1, 2, 3, And 4

Section 1

What We Learned From Four Research Attempts

What We Learned From Four Research Attempts I

Date: 2026-07-13

Presentation status: **CURRENT DRAFT -- not yet presented at a meeting.**

Latest revision: 2026-07-18. This main section explains the decisions in plain language. Exact commits, protocols, and evidence paths are preserved in the technical appendix and `docs/saq_research_direction_registry.md`.

1. Today: What Do We Need To Decide? I

This is not a presentation of a finished method.

- Should any of the four ideas continue as the next paper direction?
- Which evidence is strong enough to keep?
- Which attractive interpretations are not supported?
- What must a new direction prove before we invest again?

2. One-Minute Background I

SAQ compresses each database vector so search can compare many candidates quickly. Compression saves work, but can distort which candidates look close.

- **PCA means principal component analysis**: it rotates data to expose major directions of variation.
- **Recall means the fraction of true nearest neighbors found**.
- **QPS means queries per second**: higher is faster.
- Our bar is not "one metric improved." We need better quality at comparable speed, memory, and construction cost.

3. The Four-Attempt Map I

- **Attempt**: 1
  Plain-language question: Can a better rotation or smaller representation fix SAQ's error?
- **Decision**: Stop
- **Attempt**: 2
  Plain-language question: Can exact scalar training improve both compression and search?
- **Decision**: Keep as a baseline; stop as a method
- **Attempt**: 3
  Plain-language question: Did the chosen quality metric make an earlier result look worse?
- **Decision**: Keep as measurement evidence; stop as a method
- **Attempt**: 4
  Plain-language question: Can non-power-of-two choices use a fixed bit budget better?
- **Decision**: Portfolio closed

3. The Four-Attempt Map II

The purpose is research selection, not presenting every experiment as a win.

4. Attempt 1: The Intuition I

Imagine viewing the same cloud of points from another angle. A better angle might put the important variation into the parts SAQ represents most accurately.

Attempt 1 tested two versions:

- rotate all dimensions differently; or
- keep fewer dimensions and summarize what was removed.

A full-dimensional rotation preserves exact geometric distances. Removing dimensions does not, so the second version faced a stricter correctness test.

5. Attempt 1: Strongest Evidence And Stop I

- A promising signal on one benchmark dataset (GIST) did not repeat on a second dataset (CIFAR).
- The deliberately favorable reduced-dimension test still ranked worse than native full-dimensional SAQ before we built a search index.
- Therefore there is no general “better PCA” or “safe smaller SAQ” result.

What remains: a useful warning that one-dataset estimator gains may not replicate. What stops: both proposed mechanisms.

6. Attempt 2: The Intuition I

A scalar codebook replaces numbers with a small set of representatives. Ordinary training can settle at a locally good answer; exact training finds the best answer for the frozen one-dimensional histogram.

The question was whether that cleaner offline fit also improves actual nearest-neighbor search, after counting its training cost.

7. Attempt 2: Strongest Evidence And Stop I

- Exact training reduced reconstruction error—the difference between original and compressed values—in some regimes.
- Search quality did not improve consistently; the direction of the effect changed across settings.
- The inner exact one-dimensional optimizer is established prior work, so it is not our novelty.

What remains: a stronger offline baseline and evidence that lower compression error does not guarantee better search. What stops: presenting exact scalar training as a new database method.

8. Attempt 3: The Intuition I

Recall asks whether the returned item identifiers exactly match the true top items. A distance-ratio score instead asks how much farther the returned items are than the true ones.

Two methods can have different Recall even when their returned distances are similar. Attempt 3 checked whether this changed an earlier decision.

9. Attempt 3: Strongest Evidence And Stop I

- At one GIST operating point, the distance-based interpretation differed from the exact-identifier Recall interpretation.
- Two control settings on a second dataset remained negative.
- No new encoding or search mechanism was introduced.

What remains: evidence that the conclusion can depend on the chosen quality metric. What stops: claiming a replicated method advantage from one changed interpretation.

10. Attempt 4: The (3,5) Versus (4,4) Example I

Suppose two factors must share one fixed word with at most 16 joint states.

- Power-of-two choices naturally allow `(4,4)`: `4 × 4 = 16`.
- Allowing arbitrary positive integers also allows `(3,5)`: `3 × 5 = 15`.
- The extra choice could fit uneven data better without increasing stored bits.

This proves only that the feasible choice set is larger. It does not prove better search quality or lower system cost.

11. Attempt 4: Strongest Evidence And Stop I

- The exact frozen construction pipeline was correct on its registered synthetic checks.
- The original registered cost study projected about `24.17` CPU-hours, above its preregistered 24-hour ceiling, so it stopped.
- The later recovery attempt produced no scientific or performance result. Its last run started but ended without a valid completion record, so its runtime outcome remains unknown.
- The project decided to stop recovering the unresolved START-only experimental pipeline (`STOP_NO_FURTHER_ARTIFACT_RECOVERY` in the registry).

What remains: a mathematical possibility and a cost-stop record. What stops: further recovery work and every method, search, or performance claim.

12. Common Lesson Across The Four Attempts I

A better internal objective is not automatically a better search system.

The recurring failure modes were:

- a signal did not replicate;
- discarded information mattered;
- offline fit did not predict retrieval;
- the metric changed interpretation but not mechanism; or
- exact construction and evidence cost too much.

13. What We Should Keep I

- SAQ as a strong baseline.
- Exact scalar training as a tougher offline comparison.
- Both Recall and distance-quality views when they answer different questions.
- Negative results with frozen thresholds and fair controls.
- Early cost checks before building full experiment machinery.

These assets improve the next study even though none is a new method.

14. What We Cannot Claim I

Unsafe interpretation:

`PCA is universally optimal, exact training never helps search, or arbitrary cardinalities cannot help.`

Safe interpretation: the tested mechanisms did not produce a general, overhead-controlled improvement. The later Attempt 4 work has no valid final execution record, scientific decision, performance result, or production integration result.

15. Entry Gate For The Next Project I

Before opening another implementation branch, require:

1. a clear problem that holds on at least two independent strong baselines;
2. the most relevant published work, and why our idea is more than a direct combination;
3. one small, cheap test that can quickly show the idea is not worth pursuing, with inputs and the stop threshold fixed in advance;
4. construction, memory, and query overhead counted from the start; and
5. a fair path to moving quality versus speed, not only an internal objective.

No next Attempt 4 research or execution step is authorized.

16. Two Questions For Discussion I

1. Do we agree to close these four method lines while retaining their baselines and negative evidence?
2. For the next review, should we first select a new problem from primary work, with no implementation until a small, low-cost stop test is approved?

Section 2

Technical Appendix

Technical Appendix I

Normal presentation ends here. Use the following appendix only for questions about exact definitions, protocols, commits, or experiment details.

Section 3

Post-SAQ Pivot: Attempts 1, 2, 3, And 4

Post-SAQ Pivot: Attempts 1, 2, 3, And 4 I

PCA Replacement, Lossy Projection, Exact Scalar-Codebook DP, Recent ASQ Prior-Art Audit, Distance-Quality Re-evaluation, And An Arbitrary-Cardinality Fixed-Rate Quantization Gate

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Presentation status: **CURRENT DRAFT -- not yet presented at a meeting.**

Latest revision: 2026-07-18. Incorporates the 2026-07-13 audit of White and Singal, arXiv:2606.00289v1 and its pinned official code, the terminal A4-1S snapshot `saq-arbitrary-cardinality-analysis@f1b464b`, and the corrected A4 V2 source plus terminal PAR snapshot `saq-arbitrary-cardinality-feasibility-v2@30dfada`, independently reviewed at `fd5367e`, and the

Post-SAQ Pivot: Attempts 1, 2, 3, And 4 II

later exact executable-identity source repair `@e48df452`, independently reviewed at `c401dae`, the cache/staging disposition target `@56210f8` and review record `@249d5b8`, and the generic cache-policy source/static target `@c33a2bf`, independently reviewed at `@5db3025`, and the exact

PREP authorization target `@e7f940e`, independently reviewed at `@16a8201`, followed by the host-rebind erratum target `@5a47fed`, independently reviewed at `@b89dabe`, and the exact HOST-I-AUTH target `@212a67b`, independently reviewed at `@71e6bec`, followed by the

source-authority erratum target `@6fe8544`, independently reviewed at `@9fa9528`, and the subsequent HOST-I source-static target `@4602585`, whose direct-child independent review at `@e8e9c79` recorded one HIGH and a terminal source-static target review failure, followed by the bounded HOST-I-R1

Post-SAQ Pivot: Attempts 1, 2, 3, And 4 III

correction-only repair protocol target @ddfef99, independently reviewed at @b1a7429 with HOST_I_R1_CORRECTION_ONLY_REPAIR_PROTOCOL_REVIEW_PASS, and the exact correction target @e17f887, independently reviewed at @e10bde7 with HOST_I_R1_CORRECTION_ONLY_REPAIR_REVIEW_PASS, followed by the bounded source-history epoch-correction protocol target @dcaed57, independently reviewed

at @e98a3e4 with SOURCE_HISTORY_EPOCH_CORRECTION_PROTOCOL_REVIEW_PASS. The first authorized source-history implementation then stopped before target because its frozen ten-path closure was unsatisfiable. Correction-only path-closure erratum target @150e5b3, independently reviewed at @7749491, reached SOURCE_HISTORY_I_R1_PATH_CLOSURE_ERRATUM_REVIEW_PASS and freezes only an eight-path/five-derived-object

Post-SAQ Pivot: Attempts 1, 2, 3, And 4 IV

correction boundary. The subsequently authorized exact correction target @e9b5c83, independently reviewed at @a1198c4, reached SOURCE_HISTORY_EPOCH_CORRECTION_SOURCE_STATIC_REVIEW_PASS with zero LOW-or-higher findings. Fresh dormant PREP authorization target @cd1757e, independently reviewed at @eb1b893, then reached PREP_AUTHORIZATION_EXACT_TARGET_REVIEW_PASS with zero

LOW-or-higher findings. Its subsequently authorized unique probe returned eb1b893, and the unique invocation was consumed but left only durable START plus empty captures before the PREP body. Frozen status is therefore CRASH_OR_UNKNOWN / NO_PAR_AUTHORITY / NO_SCIENTIFIC_DECISION. Additive crash-record protocol target

Post-SAQ Pivot: Attempts 1, 2, 3, And 4 V

@1982865, independently reviewed at @8bec546, reached START_ONLY_CRASH_RECORD_PROTOCOL_REVIEW_PASS; the actual terminal record, retry, and source repair remain unauthorized. Audited closure @3577edd now records portfolio state CLOSED / STOP_NO_FURTHER_ARTIFACT_RECOVERY without changing START-only runtime status or creating an artifact, scientific,

or performance result.

Audience assumption: familiar with vector search and vector quantization at a high level, but not with SAQ's transform, segmentation, or the experiments in these repositories.

Deck status: Attempts 1, 2, and 3 are complete studies. Attempt 4 preserves the terminal A4-1S NO_GO_EXACT_SOLVER_COST, while a new primary-source review supports a narrower A4 V2 protocol for a differently defined synthetic comparative-instrument cost question. Its

Post-SAQ Pivot: Attempts 1, 2, 3, And 4 VI

additive PAR-report timing erratum is `PROTOCOL_ERRATUM_INDEPENDENT_REVIEW_PASS`. Source-only A4-V2-I reached `SOURCE_IMPLEMENTED_STATIC_REVIEW_PASS`, but the later authorized PAR command stopped before build at frozen status `ARTIFACT_INVALID`. This is an artifact- validation failure with no valid PAR authority and

no scientific decision. A4-V2-I-R1 subsequently repaired only the CPython identity source and reached `SOURCE_REPAIR_STATIC_REVIEW_PASS`; it did not run build or parity. The later CACHE-P exact target reached only `EXACT_TARGET_INDEPENDENT_REVIEW_PASS`: its protocol retains the old quarantine, requires

staged sanitized remote isolation, and rejects direct PAR-R1. CACHE-I then reached only `GENERIC_CACHE_POLICY_SOURCE_STATIC_REVIEW_PASS` for 37 filesystem sources and 38 statically enumerated executable units. A separate `compile()`-only syntax check accepted the exact locked Python snapshots without importing or executing

Post-SAQ Pivot: Attempts 1, 2, 3, And 4 VII

them. The later three-path PREP authorization target and direct- child review reached only `PREP_AUTHORIZATION_EXACT_TARGET_REVIEW_PASS`. Before PREP, static checking found that the frozen `.el9_8` Python leader had been replaced by `.el9_8.2`; PREP was not invoked and START

was not created. The additive host-rebind erratum target/review reached only `HOST_IDENTITY_REBIND_PROTOCOL_REVIEW_PASS`: host identity is not rebound, the old invocation authority is nontransferable, and the old probe is superseded and must never be reused. The later HOST-I-AUTH target/review reached

only `AUTHORIZATION_EXACT_TARGET_REVIEW_PASS`; no five-source or seven-derived-object rebind occurred, host identity is still not rebound, and no Python or PREP ran. Its later HOST-I instruction stopped before any source target because the preserved seven-component authority closure could not admit

Post-SAQ Pivot: Attempts 1, 2, 3, And 4 VIII

the required eighth component. The additive source-authority erratum target/review reached only `HOST_I_SOURCE_AUTHORITY_ERRATUM_REVIEW_PASS`. A subsequently authorized 14-path source-static target changed the five registered sources and seven derived authority objects without executing Python, build, PREP, or data access. Its

exact direct-child review failed with one HIGH because the binding and crosswalk cite a false SHA-256 for the governing erratum review. The terminal status is `SOURCE_STATIC_TARGET_REVIEW_FAIL_AUTHORITY_IDENTITY_MISMATCH` and the failed target itself remains immutable. The later correction-only repair protocol

target/review `ddfef99/@b1a7429` passed with zero LOW-or-higher findings. Exact five-path R1 target/review `e17f887/@e10bde7` then passed with zero LOW-or-higher findings after reproducing the four artifact substitutions, unchanged source/authority closure, and all 13 registered host objects. `HOST_IDENTITY_REBOUND` is established only for

Post-SAQ Pivot: Attempts 1, 2, 3, And 4 IX

that registered 13-object bundle; whole-host identity is unestablished. Source-history protocol target/review `dcaed57/@e98a3e4` then passed with zero LOW-or-higher findings, freezing a strict four-legacy/six-active epoch rule and rejecting adaptive or either-epoch matching. The first authorized `A4-V2-SOURCE-HISTORY-I` stopped before target because

required runtime-schema and maximal-witness changes had no legitimate semantic, output, or ledger delta. Path-closure erratum target/review `150e5b3/@7749491` then passed with zero LOW-or-higher findings, freezing a necessary-and-sufficient eight-path/five-derived-object cascade while preserving the two unchanged runtime artifacts exactly. Exact R1 target/review

`e9b5c83/@a1198c4` then passed static review with zero LOW-or-higher findings after changing only the one verifier source plus the five derived authority objects and two instruction files. The reviewer reproduced the five legacy overrides, strict four-legacy/six-active role partition, sole

Post-SAQ Pivot: Attempts 1, 2, 3, And 4 X

ninth protocol component, and canonical 37-source tree while the runtime schema and maximal witness remained byte-identical. This corrects the known source-history rule only in reviewed static source; no Python, syntax/import, build, verifier, PREP, data, or scientific execution occurred, so

cache-verifier/PAR readiness remains unestablished. Fresh R1 PREP authorization/review remained documentation governance. Its one later authorized probe/invocation was consumed and stopped at START-only `CRASH_OR_UNKNOWN`, before the PREP body and without scientific evidence. Reviewed crash-record protocol `1982865/@8bec546` closes only

the missing future record topology. Actual CRASH-I, source repair, retry, CACHE-BIND, PAR-R1, SRUN, real-base reads, and SAQ integration remain unauthorized.

1. Meeting Goal I

This is not a presentation of a finished method.

The goal is to explain four research attempts in enough detail to decide what they teach us, what they rule out, and what evidence is still missing:

```
Attempt 1: Is PCA a practical limitation of SAQ?
           1A. Replace full-D raw-data PCA with another isometric transform.
           1B. Physically project D -> d and summarize the omitted tail.

Attempt 2: Does histogram-exact scalar-codebook optimization improve the
           quantization and retrieval result beyond Lloyd training, and what
           remains after recent ASQ and exact-1D-DP prior work?
```

1. Meeting Goal II

Attempt 3: Does paper-exact distance quality change conclusions based on
exact-identifier Recall@k?
Attempt 4: Can arbitrary integer scalar cardinalities use a fixed B_g-bit
group word more effectively than power-of-two cardinalities while
retaining one random-access word and one lookup per group?

Main message:

Attempts 1--3 found useful local evidence, but none established a new method. For Attempt 2, the local ANN audit remains informative, while the inner exact DP primitive is established prior work and cannot be claimed as the contribution. Attempt 4's A4-1S formulation is a completed synthetic pipeline cost falsification study, not a natural-data, ANN, or systems result. A4 V2 preserves that stop and preregisters a new cost/evidence boundary. Its reporting-closure erratum is reviewed protocol evidence, not a feasibility

1. Meeting Goal III

result. Its corrected source and manifest passed independent static review, but the one authorized PAR invocation rejected the `/bin/python` symlink before any B/P worker. No build or parity fixture ran, so this adds no quantization evidence. A later exact source repair passed static review only; it has not been executed and does not change the terminal PAR result. CACHE-P then froze a governance-only isolation protocol after reviewing cache semanti and frozen source without opening the quarantine. It rejects direct PAR-R1.

CACHE-I later implemented and statically closed only that generic governance layer; its review and syntax-only check add no parity, cost, or scientific evidence. A later exact PREP authorization record passed independent review, but that is authorization-document governance only and authorizes no clone, preparation, or execution. The subsequent host-rebind protocol and HOST-I-AUT reviews likewise remain documentation governance: they rebind neither source nor host and authorize no Python or PREP. The attempted HOST-I source edge wa

1. Meeting Goal IV

stopped before edits on a seven-versus-eight contract contradiction; its reviewed additive erratum repairs only that source-authority contract and enabled a later source-static target. That target's direct-child review found one HIGH authority-identity mismatch, so that target remains failed evidence. A later correction-only target passed exact review and establishes rebound only for the registered 13-object bundle. It ran no execution and does not establish whole-host, PREP, cache-verifier/PAR, or scientific readiness. A

later reviewed source-history protocol specifies a fail-closed two-epoch repair. Its first implementation authorization stopped before target on an unsatisfiable ten-path projection; the reviewed path-closure erratum freezes the corrected eight-path boundary. The separately authorized exact R1 target then implemented only that static correction and passed direct-child review; it did not run the verifier or establish downstream readiness.
Fresh dormant PREP authorization target/review `cd1757e/@eb1b893` then passed

1. Meeting Goal V

`exact direct-child documentation review with zero LOW-or-higher findings. The later unique immediate probe passed and the unique invocation was consumed, but it stopped at durable START-only `CRASH_OR_UNKNOWN` before the PREP body. Crash-record protocol target/review `1982865/@8bec546` subsequently passed; it forms governance for a future terminal record, not execution evidence.`

Speaker notes:

- The purpose is research selection, not presenting every experiment as a win.
- Attempt 1 belongs to the SAQ repository.
- Attempt 2 comes from the sibling `vectordb` repository and tests a more classical scalar-quantization premise. The new related-work audit separates the local two-DP pipeline from the already-established inner optimizer.
- Attempt 3 returns to SAQ and asks whether Recall understated the geometric quality of any previously rejected result set.

1. Meeting Goal VI

- Attempt 4 is problem-first and fixed-rate. A4-1S passed implementation parity but failed its pipeline-cost gate. A4 V2 has a reviewed parent protocol and additive timing-closure erratum. Its PAR attempt failed only at executable-identity admission. I-R1 repairs that source

  path but remains unexecuted. CACHE-P selects staged sanitized remote isolation and rejects direct PAR-R1. CACHE-I closes its generic source/schema layer under static review only; none of these facts is quantization-quality, natural-data, ANN, or systems evidence. The later PREP
- authorization record also passed only documentation review. A subsequent static host mismatch prevented PREP and START; the reviewed host-rebind erratum is protocol governance only, and the old delayed probe is superseded and may never run or be reused.

  The later HOST-I-AUTH review freezes only the contract and projection for a possible future source target; it grants no HOST-I authority and performed no source/authority rebind, Python, or PREP. That attempted source target then stopped before edits on

1. Meeting Goal VII

- an exact seven-versus-eight component contradiction. The reviewed source-authority erratum admits the eighth component. The later 14-path HOST-I source-static target was created, but its exact review failed with one HIGH because two authority documents bind the wrong governing-review SHA-256.
  A later reviewed five-path R1 correction repaired only that provenance defect and passed independent review across all 13 registered host objects. This is bounded bundle identity, not whole- host or execution readiness. The later source-history protocol passed independent
- review, but its first implementation authorization stopped before target on an unsatisfiable ten-path closure. The reviewed path-closure erratum froze a strict eight-path/five-derived-object repair. The later exact R1 target passed direct-child static review and corrects that known source rule

1. Meeting Goal VIII

only; no verifier, PREP, cache/PAR readiness, or scientific claim follows. Fresh dormant PREP authorization target/review `cd1757e/@eb1b893` subsequently passed exact documentation review. Its unique probe/invocation was later consumed and left only START plus empty captures, so the status is

- `CRASH_OR_UNKNOWN / NO_SCIENTIFIC_DECISION`. Crash-record protocol `1982865/@8bec546` passed review, but CRASH-I, repair, retry, CACHE-BIND, and PAR-R1 remain unauthorized.

2. Current Research Position I

Project-level decision before this deck:

```
SAQ-centric incremental method search: STOP
SAQ as a strong baseline:              RETAIN
Attempt 4 A4-1S formulation:           TERMINAL COST STOP
Attempt 4 A4 V2 PAR:                   ARTIFACT_INVALID; NO SCIENTIFIC DECISI
Attempt 4 A4 V2 I-R1:                  SOURCE_REPAIR_STATIC_REVIEW_PASS ONLY
Attempt 4 A4 V2 CACHE-P target:        EXACT_TARGET_INDEPENDENT_REVIEW_PASS
Attempt 4 A4 V2 CACHE-I:               GENERIC_CACHE_POLICY_SOURCE_STATIC_REV

Attempt 4 A4 V2 PREP authorization:    PREP_AUTHORIZATION_EXACT_TARGET_REVIEW
Attempt 4 A4 V2 HOST-P erratum:        HOST_IDENTITY_REBIND_PROTOCOL_REVIEW_P
Attempt 4 A4 V2 HOST-I-AUTH:           AUTHORIZATION_EXACT_TARGET_REVIEW_PASS
Attempt 4 A4 V2 HOST-I-ERRATUM:        HOST_I_SOURCE_AUTHORITY_ERRATUM_REVIEW
Attempt 4 A4 V2 host rebound:          ESTABLISHED FOR REGISTERED 13-OBJECT E
```

2. Current Research Position II

```
Attempt 4 A4 V2 HOST-I:                SOURCE_STATIC_TARGET_REVIEW_FAIL_AUTHO
Attempt 4 A4 V2 HOST-I-R1-P:           HOST_I_R1_CORRECTION_ONLY_REPAIR_PROTO

Attempt 4 A4 V2 HOST-I-R1:             HOST_I_R1_CORRECTION_ONLY_REPAIR_REVIE
Attempt 4 A4 V2 SOURCE-HISTORY-P:      SOURCE_HISTORY_EPOCH_CORRECTION_PROTOC
Attempt 4 A4 V2 SOURCE-HISTORY-I:      STOPPED PRE-TARGET; TEN-PATH CLOSURE U
Attempt 4 A4 V2 SOURCE-HISTORY-I-R1-P: SOURCE_HISTORY_I_R1_PATH_CLOSURE_ERRAT
Attempt 4 A4 V2 SOURCE-HISTORY-I-R1:   SOURCE_HISTORY_EPOCH_CORRECTION_SOURCE
Attempt 4 A4 V2 CACHE-PREP-R1-AUTH:    PREP_AUTHORIZATION_EXACT_TARGET_REVIEW
Attempt 4 A4 V2 PREP-ISO-R1:           CRASH_OR_UNKNOWN / START_ONLY / NO SCI
```

2. Current Research Position III

```
Attempt 4 A4 V2 PREP-ISO-R1-CRASH-P:   START_ONLY_CRASH_RECORD_PROTOCOL_REVIE
Attempt 4 A4 V2 CRASH-I/BIND:          NOT AUTHORIZED / NOT RUN
Attempt 4 A4 V2 cache/PAR readiness:   NOT ESTABLISHED; CORRECTED SOURCE NOT
Attempt 4 A4 V2 direct PAR-R1:         NOT AUTHORIZED / NOT RUN
Attempt 4 A4 V2 SRUN/data:             NOT AUTHORIZED / NOT RUN
Attempt 4 A4 V2 portfolio:             CLOSED / STOP_NO_FURTHER_ARTIFACT_RECO
SAQ/index/search integration:          NOT AUTHORIZED
```

Why still present these attempts:

1. Attempts 1--3 test three intuitive claims that may arise in a meeting.
2. They distinguish optimization guarantees from retrieval guarantees.
3. They provide quantitative stop evidence instead of relying on intuition.
4. Attempt 3 tests whether an evaluation choice, rather than a quantizer mechanism, caused one earlier negative conclusion.

2. Current Research Position IV

5. Attempt 4 shows both a preregistered synthetic cost stop and how a later protocol must preserve that result while redefining its FOM and evidence boundary before any new authorization.

Unsafe interpretation:

```
PCA is universally optimal, or exact DP never helps ANN.
```

Safe interpretation:

```
The specific PCA-replacement, lossy-projection, and histogram-DP mechanisms
tested here do not supply a sufficiently general, overhead-controlled method.
The 2026 ASQ paper and pinned public code do not expose an integrated Attempt
pipeline, but the paper and earlier exact-1D-clustering work remove the inner
DP as a novelty claim.
Paper-exact distance quality changes one GIST operating-point interpretation,
but does not establish a new mechanism or a replicated method advantage.
```

2. Current Research Position V

For Attempt 4, allowing positive-integer cardinalities only enlarges the dyadic feasible set for a factorized reconstruction objective. Mixed-radix addressing and non-power-of-two scalar alphabets are prior art, and imply neither ANN improvement nor a systems advantage. A4 V2 only asks whether a complete four-arm synthetic comparative instrument fits an internal cost cap; the old `5/2` real-data projection does not transfer.

3. Attempt Map And Evidence Status I

- **Attempt**: 1A. Full-D transform replacement
  Core question: Does residual PCA or another isometric basis improve SAQ estimator behavior over raw-data PCA?
- **Main regimes**: GIST sample50k, then preregistered CIFAR60K replication
  Decision / current status: Closed after replication failure
- **Attempt**: 1B. Lossy `D -> d` projection
  Core question: Can a PCA head plus a compact tail surrogate beat native full-D SAQ?
- **Main regimes**: GIST sample50k, `960 -> 576`, favorable exact surrogate
  Decision / current status: Gate A failed; projected SAQ not built
- **Attempt**: 2. Exact scalar-codebook DP
  Core question: Does histogram-exact 1D DP improve shared dimensionwise scalar quantization over Lloyd, and is any method novelty left after prior work?
- **Main regimes**: audio, PCA CIFAR60K, PCA DEEP1M; arXiv/code audit
  Decision / current status: Stronger offline baseline; inner DP is prior art; no stable recall dominance

3. Attempt Map And Evidence Status II

- **Attempt**: 3. Distance-quality re-evaluation
  Core question: Does `1/Ratio@k` change a frozen Recall-based Pareto conclusion?
- **Main regimes**: GIST sample100k B=4; DEEP sample100k B=4/B=5 controls
  Decision / current status: Closed as metric-sensitivity evidence
- **Attempt**: 4. Arbitrary-cardinality fixed-rate words
  Core question: Does the power-of-two restriction waste material capacity at unchanged fixed payload and lookup granularity?
- **Main regimes**: A4-0 synthetic witness; terminal A4-1S correctness/cost gate; reviewed A4 V2 protocol/source, terminal pre-build PAR identity failure, static-only corrections, CACHE governance, and the consumed START-only PREP plus reviewed crash-record protocol; registered real-base gates remain unauthorized

3. Attempt Map And Evidence Status III

Decision / current status: preserve A4-1S `NO_GO_EXACT_SOLVER_COST` and A4-V2-PAR `ARTIFACT_INVALID / NO_SCIENTIFIC_DECISION`; source-history R1 passed static review only; the later unique PREP invocation stopped at `CRASH_OR_UNKNOWN / START_ONLY / NO_SCIENTIFIC_DECISION`; crash-record protocol `1982865/@8bec546` passed, but CRASH-I, repair/retry, CACHE-BIND, and PAR-R1 remain unauthorized

Speaker notes:

- Attempt 1A and 1B are related but not equivalent.
- A full-D orthogonal transform preserves exact L2 geometry.
- A physical `D -> d` projection changes the exact distance unless the omitted interaction is somehow reconstructed.
- Attempt 4 changes the feasible scalar-product cardinalities inside a fixed word; it does not yet change SAQ, CAQ, the index, or the search path.

4. Minimal SAQ Background I

SAQ compresses database vectors for approximate L2 search.

High-level pipeline:

```
raw vectors
  -> full-dimensional PCA
  -> IVF residuals
  -> contiguous 64-dimensional segments
  -> nonuniform bits across segments
  -> CAQ encoding inside each positive-bit segment
  -> progressive distance estimation during search
```

Example GIST `B=4` plan:

```
64@11 | 192@6 | 320@4 | 256@2 | 128@0
```

Meaning:

4. Minimal SAQ Background II

- **Segment**: 1
  PCA coordinates: 0-63
- **Dimensions**: 64
  Bits/dimension: 11
- **Segment**: 2
  PCA coordinates: 64-255
- **Dimensions**: 192
  Bits/dimension: 6
- **Segment**: 3
  PCA coordinates: 256-575
- **Dimensions**: 320
  Bits/dimension: 4
- **Segment**: 4
  PCA coordinates: 576-831

4. Minimal SAQ Background III

- **Dimensions**: 256
  Bits/dimension: 2
- **Segment**: 5
  PCA coordinates: 832-959
- **Dimensions**: 128
  Bits/dimension: 0

PCA is therefore not a cosmetic preprocessing step. It determines which subspaces receive high bits, low bits, or no inner-product code.

5. Common Variables I

- **Symbol**: `N`
  Meaning: number of base vectors
- **Values used here**: GIST 50,000/100,000; CIFAR 60,000; DEEP 100,000/1,000,000
- **Symbol**: `D`
  Meaning: original dimension
- **Values used here**: GIST 960; CIFAR 512; DEEP 256; audio 192
- **Symbol**: `d`
  Meaning: physically retained dimension after lossy projection
- **Values used here**: GIST 576
- **Symbol**: `K_IVF`
  Meaning: number of IVF clusters
- **Values used here**: 512 in Attempts 1 and 3
- **Symbol**: `B`
  Meaning: nominal average bits/dimension
- **Values used here**: 4 in Attempt 1; 2/4/8 in Attempt 2; 4/5 in Attempt 3

5. Common Variables II

- **Symbol**: `Q`
  Meaning: number of held-out evaluation queries
- **Values used here**: 128 or 1,000 in Attempts 1 and 3
- **Symbol**: `H`
  Meaning: equal-count histogram bins for scalar DP
- **Values used here**: default 256
- **Symbol**: `K_b`
  Meaning: scalar centroids at bitwidth `b`
- **Values used here**: up to `2^b`
- **Symbol**: `B_g`
  Meaning: fixed bits in one two-factor word for Attempt 4
- **Values used here**: 4 or 8
- **Symbol**: `S`
  Meaning: joint-state capacity of one fixed word, `2^B_g`
- **Values used here**: 16 or 256

5. Common Variables III

- **Symbol**: `K_j`
  Meaning: scalar levels assigned to factor `j`
- **Values used here**: dyadic or any positive integer
- **Symbol**: `r`
  Meaning: scalar factors grouped into one fixed word
- **Values used here**: 2 in A4-1
- **Symbol**: `w_j`
  Meaning: allocation weight for dimension `j`
- **Values used here**: 1 or historical rank-boundary weight
- **Symbol**: `R`
  Meaning: full-D orthogonal transform matrix
- **Values used here**: transform-dependent
- **Symbol**: `k`
  Meaning: number of returned neighbors evaluated
- **Values used here**: 100 in Attempt 3

5. Common Variables IV

- **Symbol**: `nprobe`
  Meaning: IVF cells scanned per query
- **Values used here**: GIST 20--400; DEEP 50--400
- **Symbol**: `d_i(q)`
  Meaning: true Euclidean distance to exact neighbor at distance rank `i`
- **Values used here**: recomputed in float64
- **Symbol**: `d_tilde_i(q)`
  Meaning: true Euclidean distance at rank `i` within the returned set
- **Values used here**: recomputed in float64

Important notation distinction:

```
K_IVF = IVF clusters
K_b   = scalar quantizer centroids
```

6. Attempt 1: Research Question I

SAQ uses raw-data PCA because it:

```
decorrelates coordinates
orders coordinates by decreasing variance
creates a low-variance tail
supports contiguous mixed-bit segmentation
```

But PCA optimizes variance/reconstruction objectives, not directly:

```
CAQ angular error
progressive-prefix distance error
top-k ranking stability
Recall-QPS-bytes
```

Attempt 1 asks two increasingly aggressive questions:

6. Attempt 1: Research Question II

```
1A. Can another full-D orthogonal transform improve SAQ while preserving L2?

1B. If low-variance tail coordinates are physically removed, can a compact
    tail summary retain enough ranking information to improve work/space?
```

Speaker notes:

- We first test whether the premise is real before training a new transform.
- A learned transform is not authorized merely because PCA optimizes a different textbook objective.

7. Attempt 1A: Transform Definitions I

Current SAQ transform:

$$T_{raw}(x) = (x - \mu_{raw})R_{raw},$$

where `R_raw` contains the eigenvectors of raw base-vector covariance.

Residual-PCA alternative:

$$r_x = x - c_{cid(x)},$$

$$T_{res}(x) = (x - \mu_{res})R_{res},$$

where `R_res` is trained from IVF residual statistics rather than raw-vector statistics.

For either full-D orthogonal matrix:

7. Attempt 1A: Transform Definitions II

$$R^T R = I,$$

so applying the same affine transform to base vectors, queries, and centroids preserves squared L2 up to float roundoff:

$$\|T(q) - T(x)\|_2^2 = \|q - x\|_2^2.$$

Controls:

```
current raw-data PCA
global residual PCA
identity coordinates
seeded random orthogonal transform
```

8. Attempt 1A Running Example I

Illustrative two-dimensional dataset:

```
raw covariance variances:       axis 1 = 100, axis 2 = 1
within-IVF residual variances:   axis 1 =   4, axis 2 = 9
```

Raw PCA ordering:

```
axis 1 first, axis 2 second
```

Residual PCA ordering:

```
axis 2 first, axis 1 second
```

If SAQ assigns more bits to earlier coordinates, the two transforms can produce different segment importance even though both preserve exact L2.

This motivates the test, but it is not evidence. The empirical question is:

8. Attempt 1A Running Example II

`Does residual ordering improve the actual packed SAQ estimator and ranking, after plan, bytes, and internal rotations are controlled?`

9. Attempt 1A Phase 1 Protocol: GIST I

- **Item**: Dataset
  Frozen value: GIST sample50k, `N=50,000`, `D=960`
- **Item**: IVF
  Frozen value: `K_IVF=512`, one recovered canonical codebook and fixed assignments
- **Item**: Budget
  Frozen value: `B=4`
- **Item**: Queries
  Frozen value: first 128 held-out queries; evaluation only
- **Item**: Fixed probes
  Frozen value: top 16 canonical IVF clusters/query
- **Item**: Fixed candidates
  Frozen value: 442,823 candidate rows/configuration
- **Item**: Ranking cutoff
  Frozen value: top 100

9. Attempt 1A Phase 1 Protocol: GIST II

- **Item**: Transforms
  Frozen value: current PCA, residual PCA, identity, random orthogonal
- **Item**: Plan controls
  Frozen value: native, frozen-PCA, uniform `960@4`
- **Item**: Segment rotations
  Frozen value: logical seeds `0..9`, plus `off`
- **Item**: Configurations
  Frozen value: `4 transforms x 3 plans x 11 controls = 132`
- **Item**: Inference
  Frozen value: query-paired percentile bootstrap, 10,000 replicates

The runner replays production packed estimators for:

9. Attempt 1A Phase 1 Protocol: GIST III

```
variance estimate
every fast prefix
every accurate prefix
full code
```

It is a fixed-candidate estimator experiment, not an end-to-end QPS run.

10. Attempt 1A Code And Artifact Map I

Branch:

```
saq-transform-analysis
evidence checkpoint: 3d94840
```

Key code:

```
script/prepare_phase1_transform_views.py
src/phase1_transform_diagnostic.cpp
script/summarize_phase1_transform.py
script/evaluate_phase1b_gate.py
```

Primary evidence:

```
docs/saq_transform_phase1_limitation_evidence_2026_07_10.md
docs/saq_transform_phase1b_external_replication_evidence_2026_07_10.md
```

10. Attempt 1A Code And Artifact Map II

Validity controls include:

```
held-out affine-transform reconstruction
orthogonality/isometry checks
identical IVF assignments and probe lists
canonical raw-space ranking labels
identical serialized bytes for matched transform pairs
distinct internal-rotation streams
```

11. GIST Native Plans And Stage Results I

Current PCA and residual PCA select the same plan and serialized bytes:

```
64@11 | 192@6 | 320@4 | 256@2 | 128@0
serialized index = 34,201,289 bytes
```

Seed-averaged mean-query results:

- **Transform**: current PCA
  Stage: fast all
- **Logical bytes/candidate**: 124
  RMSE: 0.386347
- **Top-100 agreement**: 0.593133
  Boundary inversion: `1.9294e-2`
- **Transform**: residual PCA
  Stage: fast all

11. GIST Native Plans And Stage Results II

- **Logical bytes/candidate**: 124
  RMSE: 0.386987
- **Top-100 agreement**: 0.589813
  Boundary inversion: `1.9522e-2`
- **Transform**: current PCA
  Stage: accurate prefix 1
- **Logical bytes/candidate**: 212
  RMSE: 0.0818108
- **Top-100 agreement**: 0.904930
  Boundary inversion: `9.5544e-4`
- **Transform**: residual PCA
  Stage: accurate prefix 1
- **Logical bytes/candidate**: 212
  RMSE: 0.0813185

11. GIST Native Plans And Stage Results III

- **Top-100 agreement**: 0.906188
  Boundary inversion: `9.3807e-4`
- **Transform**: current PCA
  Stage: full
- **Logical bytes/candidate**: 508
  RMSE: 0.00168088
- **Top-100 agreement**: 0.994617
  Boundary inversion: `4.0869e-6`
- **Transform**: residual PCA
  Stage: full
- **Logical bytes/candidate**: 508
  RMSE: 0.00167530
- **Top-100 agreement**: 0.994758
  Boundary inversion: `4.1323e-6`

Residual PCA improves distance RMSE at accurate/full stages, but worsens the fast stage.

12. GIST Paired Effects And Gate I

All deltas are `residual PCA - current PCA`.

- **Stage**: fast all
  Mean-query RMSE delta: `+6.398e-4`
- **Relative effect**: worse
  Ranking interpretation: top-100 and inversion also worse
- **Stage**: accurate prefix 1
  Mean-query RMSE delta: `-4.922e-4`
- **Relative effect**: `-0.602`%
  Ranking interpretation: top-100/inversion CIs include zero
- **Stage**: full
  Mean-query RMSE delta: `-5.572e-6`
- **Relative effect**: `-0.331`%
  Ranking interpretation: no reliable top-100 or inversion gain

Additional controls:

12. GIST Paired Effects And Gate II

```
accurate-prefix RMSE improvement: 10/10 rotation seeds
rotation-off accurate RMSE:       favorable
fast prefixes:                    consistently worse
identity/random with native plan: faster-looking fast stage, much worse full
identity/random with frozen plan: substantially worse ranking
```

Phase 1 interpretation:

```
A small estimator-objective mismatch exists on GIST, but no progressive or
ranking Pareto improvement is established. Standard residual PCA already
explains the signal, so a learned transform is not yet justified.
```

The only authorized continuation was one preregistered external replication.

13. Attempt 1A Phase 1b Protocol: CIFAR60K I

The replication freezes the narrow current-PCA versus residual-PCA comparison.

- **Item**: Dataset
  Frozen value: CIFAR60K, `N=60,000`, `D=512`
- **Item**: IVF
  Frozen value: `K_IVF=512`, same recovered codebook/assignments
- **Item**: Budget
  Frozen value: `B=4`
- **Item**: Queries
  Frozen value: all 1,000 held-out queries
- **Item**: Fixed probes
  Frozen value: 16/query
- **Item**: Candidate pairs
  Frozen value: 2,130,639/configuration

13. Attempt 1A Phase 1b Protocol: CIFAR60K II

- **Item**: Ranking cutoff
  Frozen value: top 100
- **Item**: Plan
  Frozen value: ‘64@9
- **Item**: Transforms
  Frozen value: current PCA, residual PCA
- **Item**: Rotation controls
  Frozen value: 10 paired seeds plus `off`
- **Item**: Configurations
  Frozen value: `2 x 11 = 22`
- **Item**: Bootstrap
  Frozen value: 10,000 query-paired replicates

Registered success required:

13. Attempt 1A Phase 1b Protocol: CIFAR60K III

```
confidence interval in the favorable direction
at least 8/10 seeds favor residual PCA
rotation-off direction also favorable
no harm at the fast stage
```

14. CIFAR60K Replication Result I

All deltas are `residual PCA - current PCA`; lower RMSE is better.

- **Stage**: fast all
  Current RMSE: 0.13355750
- **Residual RMSE**: 0.13359980
  Relative delta: `+0.0317%` worse
- **Seeds favoring residual**: 3/10
- **Stage**: accurate prefix 1
  Current RMSE: 0.03010816
- **Residual RMSE**: 0.03010955
  Relative delta: `+0.00463%` worse
- **Seeds favoring residual**: 4/10
- **Stage**: full
  Current RMSE: 0.001108523

14. CIFAR60K Replication Result II

- **Residual RMSE**: 0.001108262
  Relative delta: `-0.0235%`
- **Seeds favoring residual**: 5/10

Confidence and controls:

```
accurate-prefix CI spans zero
full-code CI spans zero
rotation-off accurate/full directions are unfavorable
fast-stage degradation is statistically supported
top-100 and boundary-inversion intervals span zero or point worse
```

Registered decision:

```
close_one_dataset_estimator_effect
```

The GIST effect does not replicate across the second frozen spectral regime.

15. Full-D Transform Complexity And Overhead I

For a dense full-D PCA implementation:

```
covariance accumulation: O(N D^2)
eigendecomposition:      O(D^3)
stored operator:         O(D^2)
base transformation:     O(N D^2)
raw-query transformation: O(D^2) per query
```

Residual PCA additionally constructs residuals in `O(ND)`, then has the same dominant covariance/eigendecomposition scale.

Concrete CIFAR operator state:

```
(D^2 + D) float32 values
= (512^2 + 512) * 4
= 1,050,624 bytes
```

15. Full-D Transform Complexity And Overhead II

Concrete GIST dense state:

```
(960^2 + 960) * 4 = 3,690,240 bytes
```

Current and residual PCA have the same operator shape, so operator size does not explain their paired result. However, the fixed-candidate study does not support an end-to-end QPS claim because raw-query transformation was not timed inside a complete

search path.

16. Attempt 1A Decision I

`Decision: CLOSE FULL-D PCA REPLACEMENT AS A MAIN DIRECTION`

Why:

1. GIST shows only small estimator RMSE changes.
2. Fast prefixes worsen.
3. Ranking gains are not statistically stable.
4. Standard residual PCA explains the only stable GIST signal.
5. The preregistered CIFAR replication fails confidence, seed, rotation-off, and no-harm requirements.

What was deliberately not implemented:

16. Attempt 1A Decision II

```
SAQ-aware learned rotation
joint transform-plan optimization
OPQ/ITQ-style learner integration
broad transform/dataset sweep
new persisted transform format
```

Claim boundary:

`This does not prove PCA universally optimal. It says the measured premise is not strong enough to justify learned replacement development.`

17. Attempt 1B: Why Test Lossy `D -> d` Projection? I

Full-D transforms preserve dimension and therefore cannot directly reduce code work or dense query-state size.

More aggressive idea:

```
retain only the high-variance PCA head
summarize the omitted tail with a small statistic
run quantization/search in d < D dimensions
```

Potential benefit:

```
fewer dimensions transformed
fewer dimensions encoded
fewer code/factor bytes
less estimator work
```

Risk:

17. Attempt 1B: Why Test Lossy `D -> d` Projection? II

`physical projection changes original-space distances and nearest neighbors`

Related-work boundary:

`truncated PCA, LeanVec-style projection, and MRQ-style head/tail refinement already occupy the broad design space`

Therefore the first test is an intentionally favorable quality upper bound, not a projected index implementation.

18. Lossy Projection Distance Decomposition I

Split PCA coordinates into retained head and omitted tail:

$$q = (q_h, q_t), \qquad x = (x_h, x_t).$$

Original exact squared L2:

$$D_0 = \|q_h - x_h\|^2 + \|q_t\|^2 + \|x_t\|^2 - 2\langle q_t, x_t\rangle.$$

Pure-head projection:

$$D_{head} = \|q_h - x_h\|^2.$$

Exact tail-norm surrogate:

18. Lossy Projection Distance Decomposition II

$$D_{head+norm} = \|q_h - x_h\|^2 + \|q_t\|^2 + \|x_t\|^2.$$

The missing information is exactly:

$$-2\langle q_t, x_t\rangle.$$

The oracle uses exact float64 head distances and exact tail norms. It adds no head quantization error, making it strictly more favorable than a deployed projected SAQ implementation.

19. Lossy Projection Running Example I

Suppose two candidates have identical head distance and equal tail norm:

```
head distance:        1.00 for both
query tail norm^2:    0.50
base tail norm^2:     0.50 for both
```

But their tail correlations differ:

```
candidate p: <q_t, p_t> = 0.45
candidate n: <q_t, n_t> = 0.35
```

Exact tail contributions:

```
p: 0.50 + 0.50 - 2*0.45 = 0.10
n: 0.50 + 0.50 - 2*0.35 = 0.30
```

Exact distances:

19. Lossy Projection Running Example II

```
p = 1.10
n = 1.30
```

Norm-only surrogate:

```
p = 2.00
n = 2.00
```

Tail norms remove average bias but cannot distinguish candidates whose omitted tail inner products carry ranking information.

20. LP-0 Gate A Protocol I

- **Item**: Dataset
  Frozen value: GIST sample50k
- **Item**: Original dimension
  Frozen value: `D=960`
- **Item**: Retained head
  Frozen value: `d=576`, exactly at an SAQ segment boundary
- **Item**: Physical reduction
  Frozen value: 40% of dimensions removed
- **Item**: IVF
  Frozen value: `K_IVF=512`
- **Item**: Budget of native comparator
  Frozen value: `B=4`
- **Item**: Queries
  Frozen value: first 128 held-out queries

20. LP-0 Gate A Protocol II

- **Item**: Fixed probes
  Frozen value: 16/query
- **Item**: Fixed candidates
  Frozen value: 442,823
- **Item**: Ranking cutoff
  Frozen value: top 100
- **Item**: Oracle head
  Frozen value: exact float64 distance over first 576 PCA coordinates
- **Item**: Tail
  Frozen value: exact float64 norms; float32 deployed summary also measured
- **Item**: Comparator
  Frozen value: seed-averaged native full-D SAQ full-code estimator

Native GIST plan:

20. LP-0 Gate A Protocol III

```
64@11 | 192@6 | 320@4 | 256@2 | 128@0
                       ^
                    d = 576
```

Projection removes both the 256@2 segment and the 128@0 tail physically.

21. LP-0 Ranking Result I

- **Estimator**: exact projected head + exact tail norms
  Top-100 agreement: 0.992890625
- **Boundary inversion rate**: `6.85248e-6`
- **Estimator**: native full-D SAQ full code
  Top-100 agreement: 0.994617188
- **Boundary inversion rate**: `4.08687e-6`

Registered deltas, projected oracle minus native SAQ:

```
top-100 agreement delta        = -0.0017265625
one-sided 95% lower bound      = -0.002578125

boundary inversion delta       = +2.76561e-6
one-sided 95% upper bound       = +4.21688e-6
```

Both ranking gates fail. Rotation-off has the same unfavorable directions:

21. LP-0 Ranking Result II

```
agreement delta  = -0.00140625
inversion delta  = +2.70090e-6
```

This is not a quantizer implementation failure: the favorable exact surrogate itself is below native SAQ ranking quality.

22. LP-0 Error Attribution I

Pooled absolute-error summary:

- **Component**: pure head, `D_head-D0`
  Bias: -0.0339155
- **MAE**: 0.0339155
  Pooled RMSE: 0.0444650
- **Component**: exact norm surrogate, `D_head+norm-D0`
  Bias: `-3.25e-6`
- **MAE**: 0.00150736
  Pooled RMSE: 0.00248180
- **Component**: float32 tail-summary precision
  Bias: `5.37e-11`
- **MAE**: `1.05e-7`
  Pooled RMSE: `1.51e-7`

22. LP-0 Error Attribution II

- **Component**: deployed norm surrogate total
  Bias: `-3.25e-6`
- **MAE**: 0.00150736
  Pooled RMSE: 0.00248180

Interpretation:

```
tail norms remove nearly all systematic bias
float32 summary precision is negligible
remaining error comes from omitted tail inner-product variation
```

The problem is not storing the tail norm inaccurately. The problem is that one norm per vector cannot recover query-candidate tail correlation near the top-k boundary.

23. Lossy Projection Complexity And Gate Logic I

An actual dense `D -> d` projection would require:

```
stored operator: O(Dd)
base transformation: O(NDd)
raw-query transformation: O(Dd)
head quantization/search: method-dependent O(d)
tail metadata: at least O(1) per vector for norm-only design
```

But Gate A deliberately removes projected-head quantization and runtime from the quality question:

```
exact head + exact tail norms already loses to native SAQ
```

Therefore adding:

23. Lossy Projection Complexity And Gate Logic II

```
projected-head CAQ error
progressive-estimator error
projection/index metadata
query transform and preparation work
```

cannot rescue the registered quality premise without introducing a different tail mechanism. Gate B, projected index construction and Recall-QPS evaluation, was not authorized.

24. Attempt 1B Decision I

`Decision: FAIL GATE A AND STOP THE REGISTERED D=576 LINE`

Why:

1. The exact head plus exact tail-norm oracle ranks candidates worse than native full-D SAQ.
2. The missing tail inner product, not summary precision, dominates the residual error.
3. A real projected quantizer would add more error and overhead.
4. Sweeping `d`, `B`, probes, or tail rules after this registered failure would be post-hoc rescue.

What the result does not prove:

```
all lossy projection is ineffective
all dimensions d fail
all richer tail representations fail
PCA truncation is universally inferior
```

24. Attempt 1B Decision II

A future projection study would require a distinct related-work-grounded mechanism and a new protocol, not continuation of this line.

25. Attempt 1 Unified Conclusion I

- **Sub-attempt**: 1A full-D residual PCA
  Positive signal: GIST accurate/full RMSE improves 0.60%/0.33%
- **Failed requirement**: no stable ranking gain; fast harm; CIFAR replication fails
  Final status: Closed
- **Sub-attempt**: 1B lossy `960 -> 576`
  Positive signal: exact tail norms reduce projection RMSE from 0.0445 to 0.00248
- **Failed requirement**: exact favorable surrogate still ranks worse than native SAQ
  Final status: Closed

Unified research conclusion:

```
PCA is not proven universally optimal, but the measured evidence does not
support spending additional degrees of freedom on either a learned full-D
replacement or the registered norm-only lossy projection.
```

Strict-reviewer view:

25. Attempt 1 Unified Conclusion II

```
The full-D signal is small and non-replicated.
The lossy signal fails before quantization is introduced.
Neither supports a new transform method or a system-level Pareto claim.
```

Speaker notes:

- Attempt 1 is a useful example of staged falsification.
- The more aggressive variant was tested with a more favorable oracle, not a weaker implementation.

26. Attempt 2: Research Question I

Attempt 2 comes from the sibling repository:

```
/rwproject/kdd-db/kluaq/vectordb
main checkpoint: f51b487
exact-hist implementation checkpoint: 9a7026d
cross-dataset comparison checkpoint:    870d829
```

Original empirical question:

```
If each coordinate uses a scalar codebook, can exact 1D dynamic programming
produce a better codebook than Lloyd refinement, and does lower raw
reconstruction SSE improve ANN recall?
```

The 2026-07-13 related-work audit adds a distinct novelty question:

```
Does White and Singal's 2026 inner-product-aware quantization paper, together
with the exact 1D k-means literature it uses, already subsume Attempt 2?
```

26. Attempt 2: Research Question II

Short answer:

```
public artifacts: no integrated two-DP pipeline is documented or exposed
combination claim: no novelty is established from that absence
inner exact scalar-codebook DP: already established; no novelty claim remains
local value that remains: controlled baseline and objective-mismatch evidence
```

The paper was submitted on 2026-05-29, before the `vectordb` exact-hist implementation on 2026-06-26. More importantly, exact 1D quantization DP predates both by decades.

Speaker notes:

- Attempt 2 is not SAQ's segment-planner DP.
- It is also not the CAQ exact direction-code oracle from the later branch.
- "No integrated public pipeline" must not be misread as "novel as a component combination."

27. Two Different Dynamic Programs I

The `vectordb` pipeline contains two DPs.

Inner DP: scalar codebook training

For one dimension `j` and bitwidth `b`:

```
input:  one sorted raw scalar column; the trainer internally constructs its
        equal-count weighted histogram
output: K_b centroids minimizing whole-bin partition SSE, with cuts restricte
        to histogram-bin boundaries
```

Only when each raw scalar is its own bin does this equal globally minimal raw-sample 1D reconstruction SSE. After DP backtracking, raw nearest-centroid SSE is separately recomputed under the deployed encoding semantics.

Outer DP: bit allocation

Given one error value `E[j,b]` per dimension and bitwidth:

27. Two Different Dynamic Programs II

$$A[j, c] = \min_b \left(A[j-1, c-b] + w_j E[j, b] \right),$$

subject to:

$$\sum_j b_j = DB.$$

The outer DP chooses the bitwidth of every dimension under an exact total bit budget. Attempt 2 changes only the inner trainer from Lloyd to exact-hist; the outer algorithm, objective form, and weights are held fixed as part

of the evaluation pipeline. The selected bit vector is not fixed: it changes when the new trainer changes `E[j,b]`.

Novelty accounting must therefore be layer-specific:

27. Two Different Dynamic Programs III

```
inner DP: exact 1D scalar clustering / quantization is prior work
outer DP: classical discrete rate allocation / multiple-choice knapsack
local experiment: asks whether the replacement changes ANN recall
```

28. Dimensionwise Scalar-Quantization Setting I

Write the base matrix as:

$$X \in \mathbb{R}^{N \times D}.$$

Attempt 2 trains on one column at a time:

$$X[:, j] = (x_{1j}, \dots, x_{Nj}),$$

and shares the resulting codebook across all `N` base vectors. For database vector:

$$x = (x_1, \dots, x_D).$$

Each dimension has an independent codebook:

28. Dimensionwise Scalar-Quantization Setting II

$$C_{j,b} = \{c_{j,b,1}, \dots, c_{j,b,K_b}\}.$$

Base encoding:

$$\hat{x}_j = \operatorname*{argmin}_{c \in C_{j,b_j}} (x_j - c)^2.$$

The query remains float. Compressed squared-L2 scan uses:

$$\hat{d}(q, x) = \sum_j (q_j - \hat{x}_j)^2.$$

Candidate bitwidths in variable mode:

28. Dimensionwise Scalar-Quantization Setting III

```
b in {0,1,2,3,4,5,6,7,8}
total budget = D * B bits/vector
```

Historical implementation boundary:

```
unpacked uint8 codes
brute-force compressed scan
L2 only
```

This is an algorithmic diagnostic, not a QPS-comparable SAQ index.

The column-wise training direction matters for the paper comparison. The paper's formal ASQ problem instead chooses a quantization set from the coordinates of one input vector; its official code also contains a separate shared-block helper, audited later.

29. Exact-Hist Construction I

For one sorted dimension with `N` values:

```
H = min(max_histogram_bins, N)
default H = 256
```

Step 1: split sorted values into `H` equal-count bins.

For each bin, record:

```
weight
sum
sum_sq
```

29. Exact-Hist Construction II

and their prefix sums.

Step 2: any contiguous bin interval `[a,b)` has one-centroid optimum:

$$\mu(a,b) = \frac{sum(a,b)}{weight(a,b)},$$

$$SSE(a,b) = sum\_sq(a,b) - \frac{sum(a,b)^2}{weight(a,b)}.$$

Step 3: partition the `H` ordered bins into `K_b` contiguous clusters.

Step 4: backtrack split points and use each interval mean as its centroid.

Implementation detail that affects the guarantee:

29. Exact-Hist Construction III

```
The histogram DP value is not inserted directly into E[j,b].
After recovering centroids, midpoint Voronoi boundaries are rebuilt, every ra
value is encoded by nearest centroid, and the raw-sample SSE of those encodin
is recomputed and passed to the outer DP.
```

Thus the reported training SSE measures the realized raw encoding, while the global optimality claim applies only to split points between indivisible bins. The recomputation evaluates the selected centroid set; it does not reoptimize that set over raw-sample

split points.

30. Scalar DP Recurrence And Guarantee I

Define:

`F[k,r] = minimum weighted SSE for the first r bins using k centroids`

Recurrence:

$$F[k, r] = \min_{m<r} \left(F[k-1, m] + SSE(m, r)\right).$$

The implementation uses monotone split points and divide-and-conquer DP optimization to compute each layer.

Exact guarantee:

30. Scalar DP Recurrence And Guarantee II

`For fixed H equal-count bins and K_b centroids, the DP returns the minimum raw-value SSE among contiguous partitions whose split points lie only at bin boundaries. Prefix weight/sum/sum_sq statistics evaluate each allowed interval's raw SSE exactly; approximation enters through forbidden in-bin split points.`

It does not guarantee:

```
minimum SSE over all N raw values when H < N
minimum query-distance error
minimum top-k boundary inversion
maximum recall
```

If `H=N`, every bin contains one sample and the formulation becomes full-sample exact 1D k-means. That expensive configuration was not run in the recorded experiments.

Strict novelty implication:

30. Scalar DP Recurrence And Guarantee III

`The missing full-scale H=N run is a limitation of this local baseline, not an open algorithmic problem in the literature.`

The recent paper's official repository directly vendors and exposes a raw exact 1D k-means solver primitive from earlier work; no public experiment caller integrates or evaluates that primitive. Scaling the local code by removing histogram coarsening would therefore

close an implementation gap, not create a method contribution.

31. White And Singal 2026: Formal Problem I

Inner Product Aware Quantization: Provably Fast, Accurate, and Adaptive Algorithms studies a different formal quantization problem.

This is a May 2026 arXiv v1 preprint by Nathan White and Krish Singal; this deck makes no venue-acceptance claim. Its contributions are the MDV/ADV inner-product objectives, rounding-distribution results, exact/approximate algorithms, and practical ASQ acceleration, not the invention

of exact 1D k-means.

For one vector `w` and a quantization-set budget `|Q(w)| <= s`, standard stochastic quantization independently rounds each coordinate to its adjacent lower or upper point in `Q(w)`, using the unique probabilities that make the rounding unbiased. Several algorithms

then return a set of size `s`.

For an unseen input/query distribution, written as `\mathcal X` below, its Average Directional Variance is:

31. White And Singal 2026: Formal Problem II

$$\mathrm{ADV}_{\mathcal{X}}(w, Q) = \sum_i \lambda_i (w_i^{\uparrow} - w_i)(w_i - w_i^{\downarrow}), \qquad \lambda_i = \mathbb{E}_{x \sim \mathcal{X}}[x_i^2].$$

The paper also studies Maximum Directional Variance, which controls the worst coordinate variance rather than their weighted sum.

Key semantic boundary:

```
paper:     adaptive, unbiased, adjacent stochastic rounding
Attempt 2: shared, biased, deterministic nearest-centroid encoding
```

32. Related DPs, Different Interval Costs I

After jointly sorting coordinate-weight pairs so that `w_1 <= ... <= w_d` while each `lambda_i` remains attached to its original coordinate, define the paper's ADV interval cost:

$$C[j,k] = \sum_{i=j}^{k} \lambda_i (w_k - w_i)(w_i - w_j).$$

A conventional equivalent form of its quantization-set DP is:

$$G[t,k] = \min_{j<k} \left(G[t-1,j] + C[j,k] \right).$$

Here `G[t,k]` is the best cost for `t` selected quantization points ending at `w_k`, with `G[1,1]`=0; the answer is `G[s,d]`.

Attempt 2 instead uses a best-mean interval cost:

32. Related DPs, Different Interval Costs II

$$F[t,r] = \min_{m<r} \left(F[t-1,m] + SSE_{\mathrm{mean}}(m,r) \right).$$

The relationship is structural, not semantic:

- **Property**: Ordered objects
  Paper ADV/ASQ: coordinates of one vector
- **Attempt 2 inner DP**: weighted bins from one database column
- **Property**: Interval representative
  Paper ADV/ASQ: two stochastic endpoints
- **Attempt 2 inner DP**: one deterministic mean
- **Property**: Interval cost
  Paper ADV/ASQ: weighted rounding variance
- **Attempt 2 inner DP**: centroid reconstruction SSE
- **Property**: Shared skeleton
  Paper ADV/ASQ: sorted contiguous partition with Monge structure

32. Related DPs, Different Interval Costs III

- **Attempt 2 inner DP**: sorted contiguous partition with monotone splits

The paper explicitly notes that if biased quantization is allowed, the optimal scheme is 1D k-means plus nearest-cluster assignment. That observation places Attempt 2's mathematical core inside established related work.

33. The Training Axis Is Transposed I

Let X be an N x D database matrix.

- **Question**: Optimization input
  Attempt 2: column X[:,j], containing N database scalars
- **Paper formal model**: row/vector X[i,:], containing D coordinates
- **Question**: Learned set
  Attempt 2: C[j,b] for one dimension and bitwidth
- **Paper formal model**: Q(X[i,:]) for one vector
- **Question**: Sharing
  Attempt 2: all N database vectors share C[j,b]
- **Paper formal model**: different vectors may have different Q
- **Question**: Rate choice
  Attempt 2: heterogeneous b_j, exact total budget
- **Paper formal model**: one cardinality budget s, with ‘

33. The Training Axis Is Transposed II

- **Question**: Encoding/objective semantics
  Attempt 2: biased deterministic nearest-centroid SSE
- **Paper formal model**: unbiased stochastic rounding optimized for IP variance

The paper's formal vector-search limitation follows from this orientation: it requires per-vector `Q` storage. Its discussion therefore leaves adaptation to potentially dynamic multi-vector quantization as future work.

This formal difference is real, but it is not enough to establish Attempt 2 novelty: dimensionwise shared scalar codebooks and discrete rate allocation are themselves classical constructions.

34. Official-Code Audit: Stronger Adjacency I

The pinned official repository contains more than the paper's formal per-vector interface:

```
batch_row_quant(X, ...)
  one Q per row/vector; matches the formal adaptive direction

block_quant(X, m, s, ...)
  flatten all N*m values in each consecutive m-column block
  and train one shared size-s Q for that block

kmeans_wilber_1d(sorted_points, k)
  raw-point globally optimal 1D k-means implementation
  requires ascending input; the wrapper does not sort
```

Pinned source: shared-block and per-row interfaces, raw exact 1D k-means wrapper.

Consequently, `block_quant(..., m=1, ...)` has exactly the same training axis as a fixed-rate shared per-dimension codebook. However:

34. Official-Code Audit: Stronger Adjacency II

```
it calls vmix_approx for ADV rather than centroid-SSE k-means
all blocks receive the same s
there is no per-dimension error table or outer bit-allocation DP
```

The `block_quant` docstring says `exact_adv`, while the pinned implementation calls `vmix_approx`; the implementation is the evidence used here. The public repository contains no caller or experiment script that proves which path generated the paper's GloVe table,

so this deck makes no such attribution.

The paper calls its GloVe result preliminary evidence and reports Recall@100 for L2 and maximum-inner-product search at an average four bits per coordinate. It does not report a controlled Lloyd-versus-exact trainer ablation, QPS/build cost, or complete quantization-set storage

accounting. That table therefore does not answer the specific Attempt 2 end-to-end question.

35. Prior-Art And Layer-By-Layer Verdict I

- **Attempt 2 layer**: Raw exact 1D k-means optimizer
  Formal paper: Recognizes it as older related work
- **Pinned public code**: Exposes `kmeans_wilber_1d` from the reused exact-1D library
  Strict-reviewer verdict: Novelty foreclosed by Wu 1991 / GLM 2017; not a White-Singal contribution
- **Attempt 2 layer**: Shared per-dimension fixed-rate set
  Formal paper: Formal unit is one vector
- **Pinned public code**: `block_quant(m=1)` is directly adjacent but uses ASQ semantics
  Strict-reviewer verdict: Not a formal guarantee; shared scalar codebooks are not enough for novelty
- **Attempt 2 layer**: Potential diagonal query-second-moment weighting
  Formal paper: ADV supplies diagonal second moments under per-vector independent unbiased rounding

35. Prior-Art And Layer-By-Layer Verdict II

- **Pinned public code**: `block_quant` uses all-one weights; no such experiment is exposed
  Strict-reviewer verdict: High objective-level adjacency, not direct pipeline coverage or a validity proof for biased SQ
- **Attempt 2 layer**: Multi-bit error table `E[j,b]`
  Formal paper: Fixes one cardinality budget `s`; no table
- **Pinned public code**: Repeated fixed-`s` calls are an API inference, not integrated or evaluated
  Strict-reviewer verdict: Not claimed novel here; separate rate-distortion prior support is required
- **Attempt 2 layer**: Outer heterogeneous-bit DP
  Formal paper: Not presented
- **Pinned public code**: Not exposed
  Strict-reviewer verdict: Not covered by this paper, but SAQ already provides DP segmentation/bit allocation

35. Prior-Art And Layer-By-Layer Verdict III

- **Attempt 2 layer**: Controlled Lloyd/exact-hist L2 recall audit
  Formal paper: Only preliminary GloVe L2/MIPS comparison against PQ
- **Pinned public code**: No public experiment caller/configuration
  Strict-reviewer verdict: Useful local negative/baseline evidence, not a quantizer contribution
- **Attempt 2 layer**: Packed variable-rate ANN system
  Formal paper: Per-vector storage is listed as a limitation/future-work issue
- **Pinned public code**: Fixed-rate block helper and 256-code search helper only
  Strict-reviewer verdict: A possible problem layer, not an achieved result in either artifact

Timeline:

```
paper and official code:       2026-05-29
vectordb exact-hist checkpoint: 2026-06-26
```

35. Prior-Art And Layer-By-Layer Verdict IV

The stronger point is not temporal priority. The paper itself traces exact 1D k-means matrix-search DP to Wu 1991. Separately, the pinned code exposes a solver primitive from the reused 2017 exact-1D-clustering implementation, without an integrated public experiment path.

Bottom line:

```
public-artifact level: no integrated shared-codebook + heterogeneous-bit +
                       outer-allocation pipeline is documented or exposed
novelty level:         absence is not combination novelty; the inner optimize
                       is foreclosed by older prior art, and remaining system
                       layers were not built or validated by Attempt 2
```

36. Attempt 2 Code Map I

Sibling repository code:

```
src/scalar_quantizer.cpp
  BuildEqualCountHistogram(...)
  HistogramIntervalSse(...)
  ComputeExactKMeansLayer(...)
  TrainExactScalarQuantizerFromSorted(...)
  midpoint boundaries, raw nearest-centroid reassignment, raw SSE

src/dimensionwise_quantizer.cpp
  per-dimension/per-bitwidth training

src/bit_allocator.cpp
  total-budget bit-allocation DP

src/eval_dimensionwise_quantizer.cpp
```

36. Attempt 2 Code Map II

```
training, encoding, brute-force scan, recall output
```

Main evidence:

```
vectordb/docs/dimensionwise_quantizer_smoke_run_2026_06_26.md
vectordb/reports/scalar_training_exact_hist_audit_2026_06_30/README.md
vectordb/reports/scalar_training_exact_hist_audit_2026_06_30/comparison.csv
```

CLI:

```
eval_dimensionwise_quantizer ... [lloyd|exact-hist] [histogram_bins]
```

37. Attempt 2 Complexity I

For one dimension and one bitwidth after sorting:

- **Stage**: Equal-count histogram build
  Complexity: `O(N)`
- **Stage**: Interval SSE query
  Complexity: `O(1)`
- **Stage**: Divide-and-conquer histogram DP
  Complexity: approximately `O(K_b H log H)`
- **Stage**: Backtracking
  Complexity: `O(K_b)`
- **Stage**: Final raw-value SSE
  Complexity: `O(N log K_b)`

Including the one-time dimension sort:

```
O(N log N + N log K_b + K_b H log H)
```

37. Attempt 2 Complexity II

Variable mode trains candidate bitwidths `0..8`:

```
O(D * [N log N + sum_b(N log K_b + K_b H log H)])
```

The outer allocation DP has approximately:

```
time   O(D * (D B) * |bitwidth choices|)
cost rows with rolling layers: O(D B)
backpointers for reconstruction: O(D * D B)
current total memory:            O(D * D B)
```

Its optimality is conditional:

```
Given one trainer-specific raw SSE table E[j,b], fixed weights w_j, the
candidate set b in {0,...,8}, and a reachable budget, the outer DP finds the
minimum table objective. It does not make the inner codebooks or Recall
globally optimal.
```

37. Attempt 2 Complexity III

Full-sample `H=N` exact training would replace the small histogram term with approximately `O(K_b N log N)` per dimension/bitwidth. It was judged too expensive for full DEEP1M and was not evaluated.

38. Exact-Hist Running Example I

One-dimensional sorted values:

```
0, 1, 2, 10, 11, 12
```

Let:

```
H = 6 bins, one value/bin
K_b = 2 centroids
```

The DP evaluates every legal split through the recurrence. The optimum is:

```
cluster 1 = {0,1,2},  centroid = 1
cluster 2 = {10,11,12}, centroid = 11
```

SSE:

38. Exact-Hist Running Example II

```
(0-1)^2 + (1-1)^2 + (2-1)^2
+ (10-11)^2 + (11-11)^2 + (12-11)^2
= 4
```

With `H=2`, each bin would already contain three values. The same partition is available, but an optimal raw split inside a bin would not be. This is the difference between histogram exactness and raw-sample exactness.

39. Audio Result: Lloyd Versus Exact-Hist I

Dataset:

```
audio base = 53,387 x 192
audio query = 200 x 192
metric = L2
mode = variable
B = 4
H = 256
source = initial H=256 comparison run
```

- **Trainer**: Lloyd
  MSE/value: 280,023
- **R@10**: 0.9260
  R@100: 0.95275
- **Training time**: 2,004.68 ms

39. Audio Result: Lloyd Versus Exact-Hist II

- **Trainer**: exact-hist
  MSE/value: 236,478
- **R@10**: 0.9270
  R@100: 0.95675
- **Training time**: 1,864.28 ms

At this operating point:

```
MSE improvement ~= 15.5%
R@10 delta       = +0.0010
R@100 delta      = +0.0040
```

This is a positive example, but it does not establish stable recall dominance or a new scalar-quantization contribution.

40. Audio Histogram-Bin Sweep I

B=4, variable allocation, later sweep rerun:

- **Histogram bins H**: 256
  MSE/value: 236,478
- **R@10**: 0.9270
  R@100: **0.95675**
- **Training time**: 1,776 ms
- **Histogram bins H**: 512
  MSE/value: 235,292
- **R@10**: 0.9340
  R@100: 0.95615
- **Training time**: 2,418 ms
- **Histogram bins H**: 1,024
  MSE/value: 234,918

40. Audio Histogram-Bin Sweep II

- **R@10**: **0.9345**
  R@100: 0.95610
- **Training time**: 3,858 ms
- **Histogram bins H**: 2,048
  MSE/value: **234,844**
- **R@10**: 0.9335
  R@100: 0.95625
- **Training time**: 7,018 ms

Observation:

`Increasing H improves the reconstruction objective almost monotonically, but R@100 is not monotonic and training time grows substantially.`

40. Audio Histogram-Bin Sweep III

At B=8, H=256 is especially coarse because K_b=256: one centroid per histogram bin leaves little partition freedom. Increasing H improves the result, but the best MSE and best recall still occur at different bin counts.

The H=256 training time here and on the previous slide comes from separate runs; it is not an internal consistency check on timing noise.

41. PCA CIFAR/DEEP Comparison I

All rows use `H=256` and variable bit allocation. Raw SSE ratio is the reported unweighted nearest-centroid `training_sse`, `exact-hist / Lloyd`; recall delta is `exact-hist - Lloyd`.

- **Dataset**: CIFAR60K
  B: 2
- **Allocation objective**: reconstruction
  Raw SSE ratio: 0.9737
- **R@100 delta**: +0.00313
- **Dataset**: CIFAR60K
  B: 2
- **Allocation objective**: rank-boundary
  Raw SSE ratio: 0.9336
- **R@100 delta**: +0.00378

41. PCA CIFAR/DEEP Comparison II

- **Dataset**: CIFAR60K
  B: 4
- **Allocation objective**: reconstruction
  Raw SSE ratio: 1.0425
- **R@100 delta**: -0.00024
- **Dataset**: CIFAR60K
  B: 4
- **Allocation objective**: rank-boundary
  Raw SSE ratio: 0.9371
- **R@100 delta**: -0.00046
- **Dataset**: DEEP1M
  B: 2
- **Allocation objective**: reconstruction
  Raw SSE ratio: 0.9774
- **R@100 delta**: +0.00751

41. PCA CIFAR/DEEP Comparison III

- **Dataset**: DEEP1M
  B: 2
- **Allocation objective**: rank-boundary
  Raw SSE ratio: 0.9477
- **R@100 delta**: +0.00681
- **Dataset**: DEEP1M
  B: 4
- **Allocation objective**: reconstruction
  Raw SSE ratio: 0.8847
- **R@100 delta**: +0.00304
- **Dataset**: DEEP1M
  B: 4
- **Allocation objective**: rank-boundary
  Raw SSE ratio: **0.8016**
- **R@100 delta**: **-0.00215**

41. PCA CIFAR/DEEP Comparison IV

Two different effects appear:

1. CIFAR `B=4` reconstruction shows histogram exactness does not guarantee lower raw-sample SSE.
2. DEEP `B=4` rank-boundary shows that much lower unweighted raw reconstruction SSE still does not guarantee higher recall.

Important correction to the earlier deck interpretation:

```
DEEP B=4 rank-boundary allocation objective
Lloyd:      1070.81
exact-hist: 1551.93
```

The weighted outer objective worsened in that cell. Therefore it is evidence against using raw SSE as a recall surrogate, not evidence that a lower rank-boundary allocation objective failed to improve recall.

42. Why Lower Reconstruction SSE Need Not Improve Recall I

Define base-vector quantization error:

$$e_x = \hat{x} - x.$$

Only the base is quantized; the query remains float. Distance error is:

$$\Delta_x = \hat{d}(q,x) - d(q,x) = -2(q-x)^T e_x + \|e_x\|^2.$$

Scalar k-means minimizes the second-order reconstruction term aggregated over base vectors:

$$\sum_x \|e_x\|^2.$$

It does not directly control:

42. Why Lower Reconstruction SSE Need Not Improve Recall II

$$-2(q-x)^T e_x,$$

which depends on query direction and signed correlation with the quantization error.

For a positive `p` and negative `n`, exact margin is:

$$m = d(q,n) - d(q,p) > 0.$$

Estimated margin is:

$$\hat{m} = m + \Delta_n - \Delta_p.$$

Recall depends on the sign of this margin, not average reconstruction SSE.

43. Ranking Running Example I

Exact distances near the retrieval boundary:

```
positive p = 1.000
negative n = 1.010
margin     = 0.010
```

Method A has smaller but oppositely signed errors:

```
Delta_p = +0.006
Delta_n = -0.006

estimated p = 1.006
estimated n = 1.004
result: inversion
```

Method B has larger common-mode error:

43. Ranking Running Example II

```
Delta_p = +0.020
Delta_n = +0.020

estimated p = 1.020
estimated n = 1.030
result: ordering preserved
```

This is why a codebook or coupled variable-bit pipeline with lower raw reconstruction SSE can nevertheless have lower recall: ranking depends on differential error across close candidates.

44. Bit-Allocation Coupling I

Changing the scalar trainer changes the full error table:

$$E[j,b] = \text{raw nearest-centroid SSE for dimension j at bitwidth b.}$$

The outer DP then selects a different bit vector:

$$(b_1, \ldots, b_D).$$

DEEP1M `B=4`, historical rank-boundary objective:

```
Lloyd histogram:
1b:61, 2b:36, 3b:33, 4b:27, 5b:19, 6b:19, 7b:13, 8b:48

Exact-hist histogram:
1b:40, 2b:48, 3b:33, 4b:30, 5b:26, 6b:28, 7b:37, 8b:14
```

44. Bit-Allocation Coupling II

Consequently recall differences combine:

```
centroid movement
encoding-boundary movement
dimension bit reallocation
surrogate-objective mismatch
```

Given this trainer-specific table and fixed weights, the outer DP is exact over the enumerated bitwidths and reachable total budget. It does not isolate a pure inner-trainer effect: changing the trainer also changes the selected bit vector.

Historical disclosure:

```
rank-boundary weights use query/ground-truth-derived pairs.
They are evidence about objective behavior, not an admissible method under th
current query-unaware project constraint.
```

45. Attempt 2 Decision I

`Decision: KEEP EXACT-HIST AS A STRONG OFFLINE BASELINE, NOT A MAIN METHOD`

What the experiment establishes:

1. The bin-restricted DP is implemented and can materially lower reported raw reconstruction SSE after nearest-centroid reassignment.
2. It sometimes improves recall, especially at lower bit budgets.
3. Historical rank-boundary gains over reconstruction survive both Lloyd and exact-hist, so they are not only a Lloyd artifact.
4. Lower raw SSE is not a stable surrogate for ANN ranking quality.

What the related-work audit establishes:

1. The White-Singal paper and pinned public code do not document or expose an integrated shared-codebook, heterogeneous-bit, outer-allocation pipeline.

45. Attempt 2 Decision II

2. The pinned code separately exposes a directly adjacent shared-block helper and a full-raw-data exact 1D k-means solver primitive; neither is shown as an integrated or evaluated Attempt 2 pipeline.
3. Exact 1D scalar clustering is much older than either project; the formal paper explicitly recognizes that prior.

Why it is not a method contribution:

1. Split points are exact only under a chosen histogram boundary restriction.
2. Full-sample exact DP was not evaluated at full scale.
3. Lower SSE does not provide a recall guarantee.
4. Recall gains are non-monotonic across `H`, datasets, budgets, and allocation objectives.
5. The inner optimizer's novelty is foreclosed by Wu 1991 / GLM 2017; outer DP novelty is not claimed here, and SAQ already supplies direct ANN prior.
6. The historical rank-boundary outer objective is query/ground-truth-aware.

45. Attempt 2 Decision III

7. Unpacked byte codes and brute-force L2 scan do not establish a competitive packed variable-rate ANN system.

Strict-reviewer summary:

```
No integrated end-to-end pipeline is visible in the paper/public code, but
that absence does not establish combination novelty. The inner optimizer is
prior art and the unbuilt system layer remains unvalidated. This is a useful
baseline and objective-mismatch result, not a new ANN quantizer or
database-systems method.
```

46. Attempt 3: Research Question I

Recall@`k` measures exact identifier overlap. For exact top-`k` set `E_k(q)` and returned set `A_k(q)`:

```
Recall@k(q) = |E_k(q) intersect A_k(q)| / k.
```

This can penalize a geometrically near-equivalent replacement at a dense top-`k` boundary. Attempt 3 therefore asks:

```
Does paper-exact distance quality change the Pareto interpretation of any
previously frozen SAQ comparison, when both plans are evaluated on the same
search-effort grid?
```

The premise is about evaluation validity, not a new quantizer:

```
Recall:        Did we return the same identifiers?
1/Ratio:       How far are the returned vectors compared with the exact ranks?
```

47. Attempt 3 Metric: `1/Ratio@k` I

For query `q`, define:

- `d_i(q)`: true Euclidean distance to the exact neighbor at distance rank `i`, for `i=1,...,k`;
- `d_tilde_i(q)`: true Euclidean distance at rank `i` after independently sorting the `k` returned identifiers by true distance.

The metric from ANN Search: Recall What Matters is:

```
Ratio@k(q) = (1/k) * sum_{i=1}^k d_tilde_i(q) / d_i(q)
```

```
1/Ratio@k(q) = k / sum_{i=1}^k d_tilde_i(q) / d_i(q).
```

Properties under valid exact top-`k` input:

47. Attempt 3 Metric: `1/Ratio@k` II

```
0 < 1/Ratio@k <= 1
higher is better
1 means equal distance quality, even if tied identifiers differ
```

Critical semantic detail: SAQ internally reports squared L2 distances, but this metric uses Euclidean distances. The evaluator must take square roots before forming the ratios.

48. Attempt 3 Running Example I

Let `k=3`. Suppose the exact result distances are:

```
d = [1.00, 1.10, 1.20].
```

The ANN result shares only one exact identifier, but after recomputing and sorting true distances it has:

```
d_tilde = [1.00, 1.11, 1.21].
```

Then:

```
Recall@3 = 1/3 = 0.3333
```

```
1/Ratio@3
  = 3 / (1.00/1.00 + 1.11/1.10 + 1.21/1.20)
  = 0.9942.
```

48. Attempt 3 Running Example II

Interpretation:

```
identifier recovery is poor
geometric result quality is nearly exact
```

This example does not imply that Recall is wrong. It shows that the two metrics answer different scientific questions.

49. Related Work And Novelty Gate I

The metric paper already contributes:

1. the judge-free, hyperparameter-free `1/Ratio@k` definition;
2. evaluation of Annoy, SuCo, HNSW, RaBitQ, and SymphonyQG on six datasets;
3. build, memory, distance-work, classification, and RAG analyses; and
4. evidence that less search effort can satisfy a distance-quality target than an equal-valued Recall target.

At `k=100` and quality `0.95`, it reports the following average ratios between the effort required by Recall and by `1/Ratio`:

- **Method**: HNSW
  Relative distance-computation effort: `9.36x`
- **Method**: RaBitQ
  Relative distance-computation effort: `2.48x`

49. Related Work And Novelty Gate II

- **Method**: SymphonyQG
  Relative distance-computation effort: `2.38x`
- **Method**: SuCo
  Relative distance-computation effort: `1.86x`
- **Method**: Annoy
  Relative distance-computation effort: `3.22x`

Strict-reviewer constraint:

`The metric is prior work, and the paper reports that relative algorithm rankings are usually stable. Re-evaluating SAQ is not a method contribution.`

Attempt 3 was therefore authorized only as a frozen retrospective falsification study. It could clarify prior evidence, but could not fit a plan to benchmark queries or claim the metric as novelty.

50. Frozen Attempt 3 Protocol I

Stages:

```
A3-0  validate the paper-exact evaluator on deterministic fixtures
A3-1  replay the frozen GIST default-versus-fac-error comparison
A3-2  test frozen DEEP B=4/B=5 rejected plans as negative controls
A3-3  apply the preregistered cross-dataset decision gate
```

- **Variable**: base vectors `N`
  GIST A3-1: 100,000
- **DEEP A3-2**: 100,000
- **Variable**: queries `Q`
  GIST A3-1: 1,000
- **DEEP A3-2**: 1,000
- **Variable**: dimension `D`
  GIST A3-1: 960

50. Frozen Attempt 3 Protocol II

- **DEEP A3-2**: 256
- **Variable**: IVF clusters `K_IVF`
  GIST A3-1: 512
- **DEEP A3-2**: 512
- **Variable**: nominal budget `B`
  GIST A3-1: 4
- **DEEP A3-2**: 4 and 5
- **Variable**: result size `k`
  GIST A3-1: 100
- **DEEP A3-2**: 100
- **Variable**: `nprobe` grid
  GIST A3-1: 20, 50, 100, 160, 200, 220, 240, 280, 300, 320, 400
- **DEEP A3-2**: 50, 100, 200, 400

Frozen common settings:

50. Frozen Attempt 3 Protocol III

```
threads = 24
QPS repetitions = 10 per row
searcher_safe_block_min_mode = 2
searcher_vars_bound_m = 4
PCA = enabled, retaining all D dimensions
```

`nprobe` is the number of IVF cells scanned. Mode 2 is the previously fixed correctness-preserving block-min path, and `m=4` is SAQ's unchanged variance bound multiplier. Neither is a parameter of `1/Ratio`; no value was selected from Attempt

3 outcomes.

51. Evaluator And Search-Code Semantics I

Core evaluator logic on branch `saq-ratio-metric-analysis`:

```
exact = sorted(true_l2(base[groundtruth_ids], query))
returned = sorted(true_l2(base[result_ids], query))
inverse_ratio = k / sum(returned[i] / exact[i] for i in range(k))
```

The implementation in `script/evaluate_inverse_ratio.py`:

1. recomputes both sets of distances in float64;
2. independently sorts by true Euclidean distance;
3. rejects duplicate, invalid, truncated, or non-finite result rows;
4. stops explicitly if an exact distance is zero; and
5. preserves every query-level value instead of only the mean.

51. Evaluator And Search-Code Semantics II

The existing upstream `utils::get_ratio` was not used as the scientific artifact because it reports forward `Ratio`, skips tiny squared distances with an epsilon, and discards query-level values.

`src/test_qps.cpp` exports result IDs only after the timed search region. Thus the evaluator can replay exact returned sets without inflating measured QPS. Eighteen deterministic Python tests cover exact, tied-distance, reordered, squared-L2, malformed-input, provenance, and frontier cases.

52. Attempt 3 Complexity And Overhead I

Given `Q` queries, returned top-`k`, and dimension `D`:

```
recompute exact and returned distances: O(Q k D) time
sort each query's distance lists:       O(Q k log k) time
Recall and ratio aggregation:           O(Q k) time
stored result identifiers:              O(Q k)
stored per-query metric values:         O(Q)
```

No `O(QND)` exhaustive search is required when exact top-`k` identifiers already exist. Only the `2Qk` selected-vector distances are recomputed.

Separation of costs:

```
ANN query latency / QPS: unchanged; result export is outside the timer
benchmark evaluation:    O(Q k D), reported separately
index build and storage: unchanged
```

52. Attempt 3 Complexity And Overhead II

The evaluator has no learned parameters. `k` is the benchmark result size, and the `nprobe` values describe the measured search-effort curve rather than a fitted decision rule.

53. GIST Comparison: Two Frozen Global Plans I

Setting: `gist_sample100k`, `K_IVF=512`, `B=4`, `D=960`, `k=100`.

```
SAQ default: 64x11 | 192x6 | 320x4 | 256x2 | 128x0
fac-error:   192x9 | 512x4 | 256x0
```

Each `dimensions x bits` term means that every coordinate in the contiguous PCA segment receives that many code bits. Both plan strings sum to 960 dimensions and were materialized as frozen SAQ `B=4` indexes before Attempt 3. The plan string

alone is not total index-byte accounting because SAQ also stores per-segment factors.

The comparison changes only the global plan:

```
same PCA vectors and IVF assignments
same CAQ encoder and serialized index format
same distance estimator, pruning, heap, and safe-search mode
same nprobe grid, threads, and repetitions
```

53. GIST Comparison: Two Frozen Global Plans II

The fac-error plan has two positive-bit segments instead of four, so it is a plausible lower-search-work shape. Its name refers to the historical offline plan artifact; Attempt 3 neither refits nor endorses that empirical objective.

54. GIST Complete Common-Grid Result I

- **nprobe**: 20
  default R@100: 0.75957
- **fac R@100**: 0.75952
  default 1/Ratio: 0.993860620
- **fac 1/Ratio**: 0.993858359
  default QPS: 31234.2
- **fac QPS**: 36687.1
- **nprobe**: 50
  default R@100: 0.92809
- **fac R@100**: 0.92769
  default 1/Ratio: 0.998671018
- **fac 1/Ratio**: 0.998668346
  default QPS: 20104.2
- **fac QPS**: 25581.3

54. GIST Complete Common-Grid Result II

- **nprobe**: 100
  default R@100: 0.98140
- **fac R@100**: 0.98069
  default 1/Ratio: 0.999773561
- **fac 1/Ratio**: 0.999769939
  default QPS: 13825.0
- **fac QPS**: 18572.6
- **nprobe**: 160
  default R@100: 0.99036
- **fac R@100**: 0.98958
  default 1/Ratio: 0.999964379
- **fac 1/Ratio**: 0.999960461
  default QPS: 10524.5
- **fac QPS**: 14189.3

54. GIST Complete Common-Grid Result III

- **nprobe**: 200
  default R@100: 0.99132
- **fac R@100**: 0.99059
  default 1/Ratio: 0.999985301
- **fac 1/Ratio**: 0.999981452
  default QPS: 9264.9
- **fac QPS**: 12333.9
- **nprobe**: 220
  default R@100: 0.99149
- **fac R@100**: 0.99079
  default 1/Ratio: 0.999987377
- **fac 1/Ratio**: 0.999983555
  default QPS: 8777.6
- **fac QPS**: 11739.0

54. GIST Complete Common-Grid Result IV

- **nprobe**: 240
  default R@100: 0.99153
- **fac R@100**: 0.99084
  default 1/Ratio: 0.999988670
- **fac 1/Ratio**: 0.999984881
  default QPS: 8313.8
- **fac QPS**: 11073.8
- **nprobe**: 280
  default R@100: 0.99163
- **fac R@100**: 0.99091
  default 1/Ratio: 0.999991688
- **fac 1/Ratio**: 0.999987869
  default QPS: 7579.7
- **fac QPS**: 9986.9

54. GIST Complete Common-Grid Result V

- **nprobe**: 300
  default R@100: 0.99164
- **fac R@100**: 0.99091
  default 1/Ratio: 0.999992743
- **fac 1/Ratio**: 0.999988921
  default QPS: 7329.5
- **fac QPS**: 9522.1
- **nprobe**: 320
  default R@100: 0.99164
- **fac R@100**: 0.99091
  default 1/Ratio: 0.999993271
- **fac 1/Ratio**: 0.999989447
  default QPS: 7035.6
- **fac QPS**: 9086.9

54. GIST Complete Common-Grid Result VI

- **nprobe**: 400
  default R@100: 0.99164
- **fac R@100**: 0.99091
  default 1/Ratio: 0.999993271
- **fac 1/Ratio**: 0.999989447
  default QPS: 6111.4
- **fac QPS**: 7784.3

At every equal `nprobe`, fac-error is faster and slightly worse under both quality metrics. Equal search effort is therefore not the scientific comparison; the relevant question is QPS at matched quality.

55. GIST Metric-Sensitive Operating Point I

Use the freshly replayed default `nprobe=200` row as the frozen target:

```
default Recall@100:     0.991320000000
default 1/Ratio@100:    0.999985300968
default QPS:            9264.923
```

Recall conclusion:

```
maximum measured fac-error Recall = 0.99091
target 0.99132 is unreachable
decision: reject at this reference quality
```

Distance-quality conclusion:

55. GIST Metric-Sensitive Operating Point II

```
fac-error nprobe=280  1/Ratio@100 = 0.999987869279
fac-error nprobe=280  QPS         = 9986.910
quality               > default target
QPS ratio             = 1.07793x
```

This is a directly measured point, not interpolation. Linear interpolation between fac-error `nprobe=240` and 280 estimates `1.17876x` QPS exactly at the target, but this is reported only as secondary evidence.

56. GIST Query-Level Attribution I

For fac-error `nprobe=280` minus default `nprobe=200`:

- **Statistic of paired 1/Ratio@100 delta**: mean
  Value: `+2.5683e-6`
- **Statistic of paired 1/Ratio@100 delta**: median
  Value: `0`
- **Statistic of paired 1/Ratio@100 delta**: minimum
  Value: `-3.1166e-4`
- **Statistic of paired 1/Ratio@100 delta**: p01
  Value: `-1.1754e-4`
- **Statistic of paired 1/Ratio@100 delta**: p05
  Value: `-4.1993e-5`
- **Statistic of paired 1/Ratio@100 delta**: queries worse
  Value: `39.6%`

56. GIST Query-Level Attribution II

- **Statistic of paired 1/Ratio@100 delta**: queries equal
  Value: `23.0%`
- **Statistic of paired 1/Ratio@100 delta**: queries better
  Value: `37.4%`

The alternative improves the aggregate mean and its separately measured marginal lower-tail statistics, but it does not dominate query by query.

Strict interpretation:

```
supported:   the historical GIST operating-point rejection is metric-sensitiv
unsupported: fac-error is uniformly better or provides a per-query guarantee
```

This paired distribution is why the mean metric alone is insufficient for a new method claim.

57. DEEP Negative Controls I

Frozen plans:

```
B=4 default:   64x6 | 192x3     candidate: 128x4 | 128x3
B=5 default:   64x7 | 192x4     candidate: 128x5 | 128x4
```

- **B**: 4
  nprobe: 50
- **default R**: 0.94550
  candidate R: 0.92381
- **default 1/Ratio**: 0.998803057
  candidate 1/Ratio: 0.998540024
- **QPS ratio**: 1.075x
- **B**: 4
  nprobe: 100

57. DEEP Negative Controls II

- **default R**: 0.97061
  candidate R: 0.94401
- **default 1/Ratio**: 0.999686070
  candidate 1/Ratio: 0.999416524
- **QPS ratio**: 1.095x
- **B**: 4
  nprobe: 200
- **default R**: 0.97641
  candidate R: 0.94884
- **default 1/Ratio**: 0.999904123
  candidate 1/Ratio: 0.999633936
- **QPS ratio**: 1.071x
- **B**: 4
  nprobe: 400

57. DEEP Negative Controls III

- **default R**: 0.97713
  candidate R: 0.94940
- **default 1/Ratio**: 0.999930322
  candidate 1/Ratio: 0.999658856
- **QPS ratio**: 1.046x
- **B**: 5
  nprobe: 50
- **default R**: 0.95180
  candidate R: 0.94251
- **default 1/Ratio**: 0.998842845
  candidate 1/Ratio: 0.998776284
- **QPS ratio**: 1.108x
- **B**: 5
  nprobe: 100

57. DEEP Negative Controls IV

- **default R**: 0.97973
  candidate R: 0.96687
- **default 1/Ratio**: 0.999728710
  candidate 1/Ratio: 0.999659199
- **QPS ratio**: 1.092x
- **B**: 5
  nprobe: 200
- **default R**: 0.98670
  candidate R: 0.97241
- **default 1/Ratio**: 0.999948614
  candidate 1/Ratio: 0.999877534
- **QPS ratio**: 1.077x
- **B**: 5
  nprobe: 400

57. DEEP Negative Controls V

- **default R**: 0.98752
  candidate R: 0.97304
- **default 1/Ratio**: 0.999974940
  candidate 1/Ratio: 0.999903505
- **QPS ratio**: 1.043x

Neither candidate reaches its default `nprobe=200` `1/Ratio` target anywhere on the measured curve. At `nprobe=200`, 97.1% of B=4 queries and 87.3% of B=5 queries are worse. Therefore `1/Ratio` is not merely accepting every faster, lower-Recall plan: both

DEEP negative decisions remain negative.

58. Attempt 3 Decision I

- **Predeclared condition**: at least one previous conclusion changes
  Outcome: pass: GIST historical reference
- **Predeclared condition**: change survives a complete measured curve
  Outcome: pass: 11-point GIST union grid
- **Predeclared condition**: positive result appears on two datasets or independent baselines
  Outcome: fail: DEEP provides controls, not a second positive
- **Predeclared condition**: query-level tails show no hidden material loss
  Outcome: mixed: 39.6% of paired GIST queries worsen
- **Predeclared condition**: a mechanism-level gap remains after related work
  Outcome: fail: no mechanism beyond the old fac-error plan

`Decision: CLOSE_AS_METRIC_SENSITIVITY_EVIDENCE`

What survives:

1. report Recall and `1/Ratio` together in future ANN studies;

58. Attempt 3 Decision II

2. distinguish identifier-boundary loss from geometric degradation; and
3. retain GIST as a concrete metric-sensitive SAQ case and DEEP as controls.

What does not reopen:

`fac-error plan sweeps`
`query-fitted metric-aware planning`
`lossy projection or exact-hist as methods`
`high-overhead CAQ, mixed-plan, graph-prefix, or search-bound directions`

The metric paper owns the general contribution. Our evidence is a bounded SAQ case study, not an independent database-systems method.

59. Attempt 4: Research Question, Old Stop, And V2 Boundary I

Current R12 reviewed snapshot: `saq-arbitrary-cardinality-feasibility-v2@3577edd`

Authoritative project-stop decision:
`saq-arbitrary-cardinality-feasibility-v2@3577edd`:
`docs/saq_a4_v2_project_stop_no_further_artifact_recovery_2026_07_18.md`

START-only crash-record protocol target/direct-child review:
`saq-arbitrary-cardinality-feasibility-v2@1982865/@8bec546`

Fresh R1 PREP authorization target/direct-child review:
`saq-arbitrary-cardinality-feasibility-v2@cd1757e/@eb1b893`

Source-history R1 correction target/direct-child review:
`saq-arbitrary-cardinality-feasibility-v2@e9b5c83/@a1198c4`

Source-history R1 path-closure erratum target/direct-child review:
`saq-arbitrary-cardinality-feasibility-v2@150e5b3/@7749491`

59. Attempt 4: Research Question, Old Stop, And V2 Boundary II

Source-history epoch-correction protocol target/direct-child review:
`saq-arbitrary-cardinality-feasibility-v2@dcaed57/@e98a3e4`

HOST-I-R1 correction target/direct-child review:
`saq-arbitrary-cardinality-feasibility-v2@e17f887/@e10bde7`

HOST-I-R1 correction-only repair protocol target/direct-child review:
`saq-arbitrary-cardinality-feasibility-v2@ddfef99/@b1a7429`

HOST-I source-static target/direct-child review:
`saq-arbitrary-cardinality-feasibility-v2@4602585/@e8e9c79`

Source-authority erratum target/review:
`saq-arbitrary-cardinality-feasibility-v2@6fe8544/@9fa9528`

Independent HOST-I-AUTH review record:
`saq-arbitrary-cardinality-feasibility-v2@71e6bec`

59. Attempt 4: Research Question, Old Stop, And V2 Boundary III

Preserved host-rebind erratum target/review:
`saq-arbitrary-cardinality-feasibility-v2@5a47fed/@b89dabe`

Preserved PREP-authorization target/review:
`saq-arbitrary-cardinality-feasibility-v2@e7f940e/@16a8201`

Preserved CACHE-I target/review:
`saq-arbitrary-cardinality-feasibility-v2@c33a2bf/@5db3025`

Preserved CACHE-P target/review:
`saq-arbitrary-cardinality-feasibility-v2@56210f8/@249d5b8`

Preserved source-repair/review snapshots:
`saq-arbitrary-cardinality-feasibility-v2@e48df452/@c401dae`

Preserved terminal PAR result/review:
`saq-arbitrary-cardinality-feasibility-v2@30dfada/@fd5367e`

Preserved terminal A4-1S snapshot: `saq-arbitrary-cardinality-analysis@f1b464b`

59. Attempt 4: Research Question, Old Stop, And V2 Boundary IV

- **Stage**: A4-0 synthetic instrument
  Committed status: `PASS_INSTRUMENT_ONLY`
- **What it means**: formulation, tiny witness, and reference checks passed
- **Stage**: A4-1P base-only protocol
  Committed status: `FROZEN_NOT_AUTHORIZED / NOT_RUN`
- **What it means**: scientific inputs and decision rule were registered; base read remained forbidden
- **Stage**: A4-1S synthetic implementation
  Committed status: `NO_GO_EXACT_SOLVER_COST`
- **What it means**: parity passed, but the reviewed frozen construction/evidence pipeline exceeded its cost ceiling
- **Stage**: A4-V2 parent protocol
  Committed status: `PROTOCOL_READY_NOT_AUTHORIZED_FOR_EXECUTION`
- **What it means**: a new FOM and evidence boundary were independently reviewed at `f86a51d`

59. Attempt 4: Research Question, Old Stop, And V2 Boundary V

- **Stage**: A4-V2-P-ERRATUM
  Committed status: `PROTOCOL_ERRATUM_INDEPENDENT_REVIEW_PASS`
- **What it means**: an additive finite PAR-report closure fixes terminal-P receipt recursion without changing the parent scientific contract
- **Stage**: A4-V2-I
  Committed status: `SOURCE_IMPLEMENTED_STATIC_REVIEW_PASS`
- **What it means**: corrected exact source `482c401`, binding, and 35-file manifest passed three independent static-only reviews; failed parent `3a4f7c5` remains non-evidence
- **Stage**: A4-V2-PAR
  Committed status: `ARTIFACT_INVALID / REVIEWED_TERMINAL`
- **What it means**: the one authorized invocation rejected the `/bin/python` symlink before a B/P worker; no build, parity artifact, valid PAR authority, or scientific decision exists
- **Stage**: A4-V2-I-R1
  Committed status: `SOURCE_REPAIR_STATIC_REVIEW_PASS`

59. Attempt 4: Research Question, Old Stop, And V2 Boundary VI

- **What it means**: exact source `e48df452` binds leader identity to stable `/proc/self/exe` bytes and same-inode `sys.executable`; three-track review passed, but no code was imported, built, or executed
- **Stage**: A4-V2-CACHE-P target
  Committed status: `EXACT_TARGET_INDEPENDENT_REVIEW_PASS`
- **What it means**: exact target `56210f8` and review `249d5b8` freeze `GO_SANITIZED_REMOTE_ISOLATION_PROTOCOL_REQUIRED / NO_GO_DIRECT_PAR_R1`; governance only, with quarantine unread and 35-source tree unchanged
- **Stage**: A4-V2-CACHE-I
  Committed status: `GENERIC_CACHE_POLICY_SOURCE_STATIC_REVIEW_PASS`
- **What it means**: target `c33a2bf` and review `5db3025` bind 37 filesystem sources, 38 executable units, and source-tree SHA-256 `95680075...`; static artifact governance only, with no PREP or parity evidence
- **Stage**: A4-V2-CACHE-PREP-AUTH
  Committed status: `PREP_AUTHORIZATION_EXACT_TARGET_REVIEW_PASS`

59. Attempt 4: Research Question, Old Stop, And V2 Boundary VII

- **What it means**: exact three-path target `e7f940e` and direct-child review `16a8201` bound a dormant PREP contract; the later host mismatch made its invocation authority nontransferable and its unique probe superseded
- **Stage**: A4-V2-PREP-HOST-P
  Committed status: `HOST_IDENTITY_REBIND_PROTOCOL_REVIEW_PASS`
- **What it means**: target/review `5a47fed/@b89dabe` bind the current `.el9_8.2` leader and complete future rebind closure; host identity is not rebound and no Python, PREP, probe, clone, token, build, data, or scientific action ran
- **Stage**: A4-V2-PREP-HOST-I-AUTH
  Committed status: `AUTHORIZATION_EXACT_TARGET_REVIEW_PASS`
- **What it means**: exact target/review `212a67b/@71e6bec` freeze only the contract and projection for a possible later source target and grant no HOST-I authority; no five-source/seven-derived-object rebind, Python, build, PREP, data, or scientific action occurred

59. Attempt 4: Research Question, Old Stop, And V2 Boundary VIII

- **Stage**: attempted A4-V2-PREP-HOST-I
  Committed status: `STOPPED_BEFORE_TARGET`
- **What it means**: exact checking exposed an unsatisfiable seven-versus-eight protocol-component closure before any source edit, Python, or PREP
- **Stage**: A4-V2-PREP-HOST-I-ERRATUM
  Committed status: `HOST_I_SOURCE_AUTHORITY_ERRATUM_REVIEW_PASS`
- **What it means**: target/review `6fe8544/@9fa9528` admit the eighth component and freeze its exact accounting/projection only; no implementation-source/derived-authority rebind or execution occurred
- **Stage**: A4-V2-PREP-HOST-I
  Committed status:
  `SOURCE_STATIC_TARGET_REVIEW_FAIL_AUTHORITY_IDENTITY_MISMATCH`

59. Attempt 4: Research Question, Old Stop, And V2 Boundary IX

- **What it means**: target `4602585` changed exactly the registered 14 paths without execution; direct-child review `e8e9c79` found one HIGH because the binding and crosswalk cite a false governing-review SHA-256; that target remains immutable failed evidence and is not retroactively passed
  by R1
- **Stage**: A4-V2-PREP-HOST-I-R1-P
  Committed status: `HOST_I_R1_CORRECTION_ONLY_REPAIR_PROTOCOL_REVIEW_PASS`
- **What it means**: target/review `ddfef99/@b1a7429` freeze only a future exact five-path correction target, corrected identities, cascade boundary, and direct-child review projection; zero LOW-or-higher findings, no repair authority, and no execution
- **Stage**: A4-V2-PREP-HOST-I-R1
  Committed status: `HOST_I_R1_CORRECTION_ONLY_REPAIR_REVIEW_PASS`

59. Attempt 4: Research Question, Old Stop, And V2 Boundary X

- **What it means**: target/review `e17f887/@e10bde7` changed only the exact five paths, reproduced all frozen identities and 13/13 registered host objects with zero LOW-or-higher findings, and establishes `HOST_IDENTITY_REBOUND` only for that registered bundle; whole-host identity remains unestablished and no execution ran
- **Stage**: A4-V2-SOURCE-HISTORY-P
  Committed status: `SOURCE_HISTORY_EPOCH_CORRECTION_PROTOCOL_REVIEW_PASS`
- **What it means**: target/review `dcaed57/@e98a3e4` independently reproduced the exact five legacy blobs, other 32 preserved sources, and strict four-legacy/six-active role partition; it rejects either-epoch/adaptive matching and authorizes no source edit or execution
- **Stage**: first A4-V2-SOURCE-HISTORY-I
  Committed status: `STOPPED_PRE_TARGET_PATH_CLOSURE_UNSATISFIABLE`
- **What it means**: exact projection proved the required ten-path target impossible because the runtime schema and maximal witness had no authorized semantic, output, observation, or ledger delta; no source target or execution occurred

59. Attempt 4: Research Question, Old Stop, And V2 Boundary XI

- **Stage**: A4-V2-SOURCE-HISTORY-I-R1-P
  Committed status: `SOURCE_HISTORY_I_R1_PATH_CLOSURE_ERRATUM_REVIEW_PASS`
- **What it means**: target/review `150e5b3/@7749491` froze the necessary-and-sufficient future eight-path/five-derived-object cascade, preserved the two runtime artifacts exactly, and retained the original contract as the sole ninth review component; zero LOW-or-higher findings and governance only
- **Stage**: A4-V2-SOURCE-HISTORY-I-R1
  Committed status:
  `SOURCE_HISTORY_EPOCH_CORRECTION_SOURCE_STATIC_REVIEW_PASS`
- **What it means**: target/review `e9b5c83/@a1198c4` changed the exact eight paths and only one verifier source (+38 net lines), independently reproduced five legacy overrides, the strict four-legacy/six-active role partition, nine-component boundary, and canonical 37-source tree, and preserved the runtime schema/maximal witness byte-for-byte; zero LOW-or-higher findings, but no syntax/import, verifier, build, PREP, data, or scientific execution

59. Attempt 4: Research Question, Old Stop, And V2 Boundary XII

- **Stage**: A4-V2-CACHE-PREP-R1-AUTH
  Committed status: `PREP_AUTHORIZATION_EXACT_TARGET_REVIEW_PASS`
- **What it means**: target/review `cd1757e/@eb1b893` independently closed the reviewed source head, current source/authority identities, corrected source history, and registered-host ceiling with zero LOW-or-higher findings; documentation governance only, with the ordinary target-head observation distinct from the reserved immediate review-head probe
- **Stage**: A4-V2-CACHE-PREP-ISO-R1
  Committed status: `CRASH_OR_UNKNOWN / START_ONLY / NO_SCIENTIFIC_DECISION`
- **What it means**: the exact immediate probe passed and the unique invocation was consumed, but only durable START and empty captures exist; no TERMINAL, isolation root, clone, receipt, or PREP-body entry
- **Stage**: A4-V2-CACHE-PREP-ISO-R1-CRASH-P
  Committed status: `START_ONLY_CRASH_RECORD_PROTOCOL_REVIEW_PASS`

59. Attempt 4: Research Question, Old Stop, And V2 Boundary XIII

- **What it means**: target/review `1982865/@8bec546` freeze only the future reviewed terminal-record topology, with zero LOW-or-higher findings and no retry or repair authority
- **Stage**: A4-V2-CACHE-PREP-ISO-R1-CRASH-I / CACHE-BIND
  Committed status: `NOT_AUTHORIZED / NOT_RUN`
- **What it means**: no terminal record or binding was formed; portfolio closure does not reclassify START-only runtime state
- **Stage**: A4-V2-PAR-R1
  Committed status: `NOT_AUTHORIZED / NOT_RUN`
- **What it means**: no build or corrected parity event occurred
- **Stage**: A4-V2-SRUN / data
  Committed status: `NOT_AUTHORIZED / NOT_RUN`
- **What it means**: no synthetic scientific event or benchmark/base/query/index read occurred

59. Attempt 4: Research Question, Old Stop, And V2 Boundary XIV

- **Stage**: A4-V2 project portfolio
  Committed status: `CLOSED / STOP_NO_FURTHER_ARTIFACT_RECOVERY`
- **What it means**: audited closure `3577edd` stops investment in the recovery chain; it is not an artifact terminal, scientific result, or performance result
- **Stage**: SAQ/index/search integration
  Committed status: `NOT_AUTHORIZED`
- **What it means**: no production representation or query-path change may be made

Research question:

```
Under one fixed B_g-bit word for a two-factor group, can positive-integer
scalar cardinalities use the joint-state budget more effectively than
power-of-two cardinalities while retaining one random-access word and one
expanded-table lookup per group?
```

The competing factorized feasible sets are:

59. Attempt 4: Research Question, Old Stop, And V2 Boundary XV

```
dyadic:      K_j in {1, 2, 4, 8, ...}
arbitrary:   K_j in positive integers
both:        product_j K_j <= S = 2^B_g
```

This is a fixed-address coding opportunity study. It does not modify CAQ or SAQ, and it produced no natural-data, Recall, QPS, or systems claim.

60. Correct Fixed-Rate Formulation And Witness I

First correct a tempting but invalid rate comparison:

```
(3,3) has 9 joint states and needs 4 fixed bits.
It cannot be compared at equal fixed rate with a 3-bit (4,2) product.
```

The frozen equal-rate witness instead uses:

- **Quantity**: word width
  Value: `B_g=4` bits
- **Quantity**: joint capacity
  Value: `S=16` addresses
- **Quantity**: best arbitrary witness allocation
  Value: `(3,5)`, 15 used states
- **Quantity**: selected dyadic allocation
  Value: `(4,4)`, 16 used states

60. Correct Fixed-Rate Formulation And Witness II

For two factors with zero-based labels `z_1` and `z_2`, mixed-radix addressing is direct:

```
u = z_1 + K_1 z_2,          0 <= u < K_1 K_2 <= S.
```

The address is still one fixed word and the query interface is still one `S`-entry lookup per group. Unused addresses remain paid capacity; entropy or Huffman expected length is not a fixed-payload saving.

The only unconditional optimization statement is:

```
D*_arbitrary <= D*_dyadic
```

for the same exact factorized reconstruction-SSE objective, because the arbitrary-cardinality feasible set contains the dyadic set. It is not a Recall, QPS, or block-VQ dominance theorem.

61. Prior Art And Claim Boundary I

The following are known ingredients, not Attempt 4 novelty:

```
non-power-of-two scalar quantization
transform coding and bit allocation
adaptive or irregular product quantization
mixed-radix fixed addressing
finite scalar quantization and non-dyadic palette constructions
entropy-constrained and variable-length coding
```

Therefore a publishable claim would need evidence beyond the primitive:

1. a material gap on natural ANN residuals at matched fixed payload;
2. a same-capacity trained block-VQ control, because unrestricted block VQ weakly dominates a factorized codebook in reconstruction expressiveness;
3. complete training, metadata, expanded-table, packing, and scan-cost accounting; and

61. Prior Art And Claim Boundary II

4. only after a new frozen systems protocol, a replicated Recall-throughput effect against competitive baselines.

Current boundary:

```
PRESERVE A4-1S TERMINAL COST STOP
A4 V2 CORRECTED PROTOCOL AUTHORITY REVIEWED
A4-V2-I SOURCE_IMPLEMENTED_STATIC_REVIEW_PASS
A4-V2-PAR ARTIFACT_INVALID / REVIEWED TERMINAL / NO VALID PAR AUTHORITY
A4-V2-I-R1 SOURCE_REPAIR_STATIC_REVIEW_PASS ONLY
A4-V2-CACHE-P TARGET EXACTLY REVIEWED / NO DIRECT PAR-R1
A4-V2-CACHE-I GENERIC_CACHE_POLICY_SOURCE_STATIC_REVIEW_PASS ONLY
```

61. Prior Art And Claim Boundary III

A4-V2-CACHE-PREP-AUTH PREP_AUTHORIZATION_EXACT_TARGET_REVIEW_PASS ONLY
A4-V2-PREP-HOST-P HOST_IDENTITY_REBIND_PROTOCOL_REVIEW_PASS ONLY
A4-V2-PREP-HOST-I-AUTH AUTHORIZATION_EXACT_TARGET_REVIEW_PASS ONLY
A4-V2-PREP-HOST-I ATTEMPT STOPPED BEFORE TARGET ON 7-VS-8 CONTRADICTION
A4-V2-PREP-HOST-I-ERRATUM HOST_I_SOURCE_AUTHORITY_ERRATUM_REVIEW_PASS ONLY
A4-V2-PREP-HOST-I SOURCE_STATIC_TARGET_REVIEW_FAIL_AUTHORITY_IDENTITY_MISMATC
A4-V2-PREP-HOST-I-R1-P HOST_I_R1_CORRECTION_ONLY_REPAIR_PROTOCOL_REVIEW_PASS

A4-V2-PREP-HOST-I-R1 HOST_I_R1_CORRECTION_ONLY_REPAIR_REVIEW_PASS
HOST_IDENTITY_REBOUND ESTABLISHED FOR REGISTERED 13-OBJECT BUNDLE ONLY
WHOLE-HOST IDENTITY NOT ESTABLISHED
A4-V2-SOURCE-HISTORY-P SOURCE_HISTORY_EPOCH_CORRECTION_PROTOCOL_REVIEW_PASS
A4-V2-SOURCE-HISTORY-I STOPPED PRE-TARGET; TEN-PATH CLOSURE UNSATISFIABLE
A4-V2-SOURCE-HISTORY-I-R1-P SOURCE_HISTORY_I_R1_PATH_CLOSURE_ERRATUM_REVIEW_P
A4-V2-SOURCE-HISTORY-I-R1 SOURCE_HISTORY_EPOCH_CORRECTION_SOURCE_STATIC_REVIE

61. Prior Art And Claim Boundary IV

```
KNOWN SOURCE-HISTORY RULE CORRECTED IN STATIC SOURCE / VERIFIER NOT RUN
A4-V2-CACHE-PREP-R1-AUTH PREP_AUTHORIZATION_EXACT_TARGET_REVIEW_PASS ONLY
CACHE-VERIFIER AND PAR READINESS NOT ESTABLISHED
A4-V2-CACHE-PREP-ISO-R1 CRASH_OR_UNKNOWN / START_ONLY / NO SCIENTIFIC DECISIO
A4-V2-CACHE-PREP-ISO-R1-CRASH-P START_ONLY_CRASH_RECORD_PROTOCOL_REVIEW_PASS
A4-V2-CACHE-PREP-ISO-R1-CRASH-I, REPAIR, RETRY, AND CACHE-BIND NOT AUTHORIZED
A4-V2-PAR-R1 NOT AUTHORIZED / NOT RUN

A4-V2-SRUN, DATA, AND SAQ CHANGES NOT AUTHORIZED / NOT RUN
A4-V2 PORTFOLIO CLOSED / STOP_NO_FURTHER_ARTIFACT_RECOVERY
NO ARTIFACT TERMINAL / NO SCIENTIFIC DECISION / PERFORMANCE_NOT_YET_MEASURED
```

61. Prior Art And Claim Boundary V

The fixed-word opportunity remains a mathematical possibility, but the frozen exact A4-1S construction/evidence pipeline did not meet its preregistered cost ceiling. V2 does not rescue or reinterpret that result. It freezes a new synthetic comparative-instrument question and cannot

claim a new quantization primitive, feasibility, or SAQ integration. The erratum repairs protocol closure only. The later PAR attempt reached module import but failed at the leader executable's no-follow identity read, before build or parity. It adds only

artifact-validation failure evidence, not evidence about the candidate. I-R1 then repaired that dedicated identity acquisition and passed exact- commit static review, but no corrected conductor was executed. It therefore adds source-correctness evidence only and does not supersede the

PAR terminal status. CACHE-P later reviewed CPython cache semantics and exact committed source, retained the unread quarantine, and froze a staged sanitized remote-isolation authority DAG. Its exact-target review found no LOW+ issue, but its claim ceiling is

61. Prior Art And Claim Boundary VI

artifact governance only. It changes no implementation or environment, creates no clone, and rejects direct PAR-R1.

CACHE-I then added only the frozen generic cache-policy source/schema layer. Exact target `c33a2bf` and direct-child review `5db3025` close 37 filesystem sources plus one separate inline bootstrap, four schema/maximal-instance pairs, and a 38-unit static import/process/mutation inventory. A separately authorized

Python 3.9 `compile()`-only check accepted the exact locked source snapshots without importing or executing them. The maximum claim is `GENERIC_CACHE_POLICY_SOURCE_STATIC_REVIEW_PASS`, not PREP readiness, parity, feasibility, an SAQ limitation, systems performance, novelty, or a method.

The subsequent exact three-path PREP authorization target `e7f940e` and direct-child review `16a8201` reached `PREP_AUTHORIZATION_EXACT_TARGET_REVIEW_PASS` with 0 LOW+ findings. This froze only a dormant execution contract. Before invocation, static checking found that a root RPM update had replaced the

61. Prior Art And Claim Boundary VII

frozen `.el9_8` Python leader with `.el9_8.2`; PREP was not invoked, START was not created, and no runtime terminal status exists. Host-rebind erratum target/review `5a47fed/@b89dabe` then reached only `HOST_IDENTITY_REBIND_PROTOCOL_REVIEW_PASS`. It does not rebind source or establish PREP

readiness. The old invocation authority is unspent but nontransferable; its probe is unspent but superseded and must never run or be reused. The later exact HOST-I-AUTH target/review `212a67b/@71e6bec` reached only `AUTHORIZATION_EXACT_TARGET_REVIEW_PASS`. It froze a possible future fourteen-path

source edge but changed none of the five sources or seven derived authority objects. When HOST-I was later instructed, exact projection checking found that the frozen seven-component authority closure could not admit the required eighth host-rebind component, so

61. Prior Art And Claim Boundary VIII

work stopped before any source target. Additive source-authority erratum target/review `6fe8544/@9fa9528` reached only `HOST_I_SOURCE_AUTHORITY_ERRATUM_REVIEW_PASS`: it admits that eighth component and freezes exact future byte/accounting semantics, but still changes no implementation source or derived authority object. The subsequently authorized

HOST-I produced exact source-static target `4602585`. Its direct-child review `e8e9c79` found one HIGH: the binding and crosswalk contain a false digest for the governing erratum review. Therefore the status is `SOURCE_STATIC_TARGET_REVIEW_FAIL_AUTHORITY_IDENTITY_MISMATCH`, and Python/PREP remain `NOT_AUTHORIZED_NOT_RUN`.

61. Prior Art And Claim Boundary IX

Correction-only protocol target/review `ddfef99/@b1a7429` later reached `HOST_I_R1_CORRECTION_ONLY_REPAIR_PROTOCOL_REVIEW_PASS`. Exact R1 target/review `e17f887/@e10bde7` then reached `HOST_I_R1_CORRECTION_ONLY_REPAIR_REVIEW_PASS` and established rebound only for the registered 13-object bundle; whole-host identity remains unestablished. Source-history protocol target/review `dcaed57/@e98a3e4` subsequently reached `SOURCE_HISTORY_EPOCH_CORRECTION_PROTOCOL_REVIEW_PASS`, freezing an

exact role-preassigned two-epoch rule. The first implementation authorization then stopped before target because its ten-path closure was unsatisfiable. Path-closure erratum target/review `150e5b3/@7749491` subsequently reached `SOURCE_HISTORY_I_R1_PATH_CLOSURE_ERRATUM_REVIEW_PASS`, freezing only an exact eight-path/five-derived-object correction boundary. Exact R1 target/review `e9b5c83/@a1198c4` subsequently

61. Prior Art And Claim Boundary X

reached SOURCE_HISTORY_EPOCH_CORRECTION_SOURCE_STATIC_REVIEW_PASS with zero LOW-or-higher findings. It corrected only the known verifier source rule and five derived authority objects while preserving the two runtime artifacts; the verifier was not run, so all cache-verifier/PAR readiness claims remain forbidden. Fresh

R1 PREP authorization target/review cd1757e/@eb1b893 subsequently passed exact documentation review. Its later unique probe and invocation were consumed and stopped at START-only CRASH_OR_UNKNOWN before the PREP body. Reviewed protocol 1982865/@8bec546 closes only the future terminal-record topology; CRASH-I, repair,

retry, and CACHE-BIND remain unauthorized.

62. A4-0 Synthetic Instrument Result I

- **Quantity**: fixed capacity
  Observed: 16
- **Quantity**: arbitrary cardinalities / used states
  Observed: `(3,5)` / 15
- **Quantity**: arbitrary distortion per vector
  Observed: 0
- **Quantity**: dyadic cardinalities / used states
  Observed: `(4,4)` / 16
- **Quantity**: dyadic distortion per vector
  Observed: 0.1
- **Quantity**: mixed-radix LUT/direct L2 maximum difference
  Observed: 0
- **Quantity**: unrestricted 16-codeword block-VQ distortion
  Observed: 0

62. A4-0 Synthetic Instrument Result II

The constructed feasible-set gap is recovered exactly:

```
D*_dyadic - D*_arbitrary = 0.1 per vector.
```

Seven deterministic tests pass: exact scalar DP, duplicate/weighted support, both allocation modes against exhaustive enumeration, all 15 mixed-radix tuples, expanded-LUT/direct-distance parity, and the frozen witness/Huffman values.

Interpretation ceiling:

```
PASS_INSTRUMENT_ONLY
```

The unrestricted block-VQ oracle also reaches zero. No dataset, query, ground truth, PCA, IVF, or SAQ artifact was used. The result validates the reference instrument and strict dyadic-feasible-set witness, not a practical method.

63. Frozen A4-1 Base-Only Falsification Gate I

Status: `FROZEN_NOT_AUTHORIZED / NOT_RUN / FORECLOSED_BY_A4-1S`

Retained only for protocol provenance, A4-1 would have used query-unaware GIST sample50k and CIFAR60K residual panels at fixed 4-bit and 8-bit two-factor words. Its registered primary arms were:

```
dyadic_word
arbitrary_word
trained_block_vq          same joint capacity
```

A global capped dyadic pack is retained as an attribution control. It has the same 32/64-byte database payload but a different 128-scalar-lookup interface, so it cannot silently replace the matched-word comparison.

For items 1--4 below, passage requires Holm-adjusted one-sided evidence to reject the corresponding non-positive null in every dataset-by-rate cell. Items 5--6 are additional hard point-estimate gates:

63. Frozen A4-1 Base-Only Falsification Gate II

1. arbitrary words remove more than 5% of held-out dyadic error;
2. a positive trained-block-VQ opportunity exists;
3. arbitrary words close more than half of the dyadic-to-block-VQ gap;
4. the base-pair absolute-distance-error contrast is positive;
5. at least 48 of 64 groups have positive dyadic-error removal; and
6. every leave-one-group-out pooled removal estimate remains at least 5%.

The first four contrasts are frozen as a 16-hypothesis Holm family across the four dataset-by-rate cells. Every primary arm pays identical fixed payload and full expanded lookup capacity.

The registered maximum positive outcome would have been only:

`GO_TO_SYSTEMS_PREREG`

63. Frozen A4-1 Base-Only Falsification Gate III

That would have established a data-level word-local opportunity and permitted writing a later systems protocol. It still would not have established Recall, QPS, or a method claim. A4-1S did not pass, so reading the registered base inputs remains

forbidden and this gate must not be opened.

A4 V2 does not reopen A4-1. Its source-only implementation and independent static review completed, and the user separately authorized frozen PAR. That one invocation stopped at `ARTIFACT_INVALID` before any B/P worker, build, or parity fixture, so it created

no valid PAR authority and no scientific decision. The later clean I-R1 source correction and independent static review are now complete. CACHE-P subsequently reviewed the cache/staging question at exact target `56210f8`, with its independent review recorded at

`249d5b8`. It requires sanitized remote isolation and rejects direct PAR-R1, but changes no implementation or environment and leaves the quarantine unread and uncleaned. CACHE-I subsequently reached `GENERIC_CACHE_POLICY_SOURCE_STATIC_REVIEW_PASS` at target/review `c33a2bf/@5db3025`; it adds only static governance sources

63. Frozen A4-1 Base-Only Falsification Gate IV

and schemas. The later PREP authorization target/review e7f940e/@16a8201 reached only PREP_AUTHORIZATION_EXACT_TARGET_REVIEW_PASS. Static checking then found the .el9_8 -> .el9_8.2 leader replacement before PREP or START. Host-rebind erratum target/review 5a47fed/@b89dabe reached only HOST_IDENTITY_REBIND_PROTOCOL_REVIEW_PASS; the old probe is superseded and must

never be reused. HOST-I-AUTH target/review 212a67b/@71e6bec then reached only AUTHORIZATION_EXACT_TARGET_REVIEW_PASS: no source/authority rebind, Python, START, or PREP occurred. A later HOST-I instruction stopped before any source target on the exact seven-versus-eight component contradiction. Source-authority erratum target/review 6fe8544/@9fa9528 reached

63. Frozen A4-1 Base-Only Falsification Gate V

only HOST_I_SOURCE_AUTHORITY_ERRATUM_REVIEW_PASS; it changes no implementation source or derived authority object and grants no HOST-I authority. The later source-static target/review 4602585/@e8e9c79 ended at SOURCE_STATIC_TARGET_REVIEW_FAIL_AUTHORITY_IDENTITY_MISMATCH: one HIGH false governing-review digest in the binding and crosswalk, with host

rebound not established. Correction-only protocol target/review ddfef99/@b1a7429 subsequently passed with zero LOW-or-higher findings. Exact R1 target/review e17f887/@e10bde7 then corrected only that authority identity and passed independent review across all 13 registered host objects. Bundle rebound is established; whole-host

identity and execution readiness are not. Source-history protocol target/review dcaed57/@e98a3e4 passed and froze a strict future two-epoch repair. The first authorized implementation stopped before target on its unsatisfiable ten-path closure; path-closure erratum target/review 150e5b3/@7749491 then passed only for

63. Frozen A4-1 Base-Only Falsification Gate VI

a future exact eight-path/five-derived-object correction. Exact R1 target/review e9b5c83/@a1198c4 then implemented only that static repair and passed direct-child review with zero LOW-or-higher findings. The corrected verifier was not run, so cache-verifier/PAR readiness remains unestablished. Fresh dormant PREP authorization

target/review cd1757e/@eb1b893 then reached PREP_AUTHORIZATION_EXACT_TARGET_REVIEW_PASS with zero LOW-or-higher findings. Its later unique immediate probe passed and invocation was consumed, but only durable START and empty captures exist; the frozen result is CRASH_OR_UNKNOWN / NO_SCIENTIFIC_DECISION. Crash-record protocol 1982865/@8bec546 passed independent

review. CRASH-I, repair, retry, CACHE-BIND, PAR-R1, SRUN, base, and query reads remain unauthorized.

64. A4-1S Synthetic Implementation And Cost Gate I

Status: `NO_GO_EXACT_SOLVER_COST` -- committed and independently reviewed

A4-1S exists only to establish that the exact registered machinery is correct, deterministic, representation-complete, and affordable before any base read. Its committed parity phase executed:

```
1,044 exact scalar cases: 780 exhaustive + 8 bit-pattern + 256 PCG64
3,961 persisted scalar solutions and exact-rational replays
2,433,600 allocation decisions, all executed in canonical order and digested
explicit mixed-radix, B4/B8 packing, and LUT fixtures
64 deterministic trained-block-VQ cases plus five named microfixtures
six status-precedence fixtures and 22/22 deterministic tests
```

64. A4-1S Synthetic Implementation And Cost Gate II

All registered parity checks passed. The six canonical evidence files are at `335837e`; independent review at `988ace0` found zero blocker or high-severity issues. This supports `PASS_PARITY` for the instrument only.

The frozen one-panel `8192 x 128` cost command then reached its registered early-stop terminal after publishing coordinate 60, leaving 61 complete scalar-coordinate shards. The reviewed integer accounting is:

```
timed CPU                       34,805,155,525 us
frozen projection factor                 5 / 2
projected cost                 24.170246892361 CPU-hours
registered ceiling            24.000000000000 CPU-hours
excess                             612.8888125 seconds
```

64. A4-1S Synthetic Implementation And Cost Gate III

Canonical shard serialization used 18,096.207493 CPU-seconds, versus 16,706.810033 seconds for exact scalar construction. The result therefore rejects the whole frozen construction/evidence pipeline, not the scalar solver alone and not arbitrary-cardinality quantization quality.

Status precedence prevents a broken instrument from becoming scientific evidence:

- **Condition**: artifact/schema or implementation parity defect
  Outcome: `ARTIFACT_INVALID` or `IMPLEMENTATION_INVALID`
- **Condition**: trained block-control defect
  Outcome: `CONTROL_INVALID`
- **Condition**: selected scalar alphabet collapses at binary32
  Outcome: `NO_GO_REPRESENTATION`
- **Condition**: frozen exact construction/evidence pipeline cost ceiling is exceeded
  Outcome: `NO_GO_EXACT_SOLVER_COST`

64. A4-1S Synthetic Implementation And Cost Gate IV

- **Condition**: every correctness, representation, control, and cost gate passes
  Outcome: `PASS_SYNTHETIC_GATE_ONLY`

`PASS_SYNTHETIC_GATE_ONLY` would have permitted only asking the user whether to authorize the already-registered base gate. It did not occur.

Cost evidence is committed at `9ce1052`; the independent review and terminal decision are committed at `f1b464b`. Allocation, block-VQ, and encoding cost phases did not execute after the scalar-prefix early stop, so the result makes no claim

about them. A4-1 was not run, and untracked worktree files are deliberately excluded.

65. A4 V2: Terminal PAR Failure, Source Repair, And Governance Chain I

Primary review: `saq-arbitrary-cardinality-feasibility-v2@c4ccea3`

Reviewed protocol snapshot: `saq-arbitrary-cardinality-feasibility-v2@f86a51d`

Corrected protocol-authority snapshot:
`saq-arbitrary-cardinality-feasibility-v2@f13a383`

Independent erratum-review head:
`saq-arbitrary-cardinality-feasibility-v2@98e6999`

Corrected source snapshot: `saq-arbitrary-cardinality-feasibility-v2@482c401`

Independent source-review head:
`saq-arbitrary-cardinality-feasibility-v2@4ce2e69`

Reviewed PAR authorization base:
`saq-arbitrary-cardinality-feasibility-v2@2e983a0`

Terminal PAR result snapshot: `saq-arbitrary-cardinality-feasibility-v2@30dfada`

65. A4 V2: Terminal PAR Failure, Source Repair, And Governance Chain II

Independent terminal-review head: `saq-arbitrary-cardinality-feasibility-v2@fd5367e`

I-R1 source-repair authorization: `saq-arbitrary-cardinality-feasibility-v2@fe10bc3`

Exact source-repair snapshot: `saq-arbitrary-cardinality-feasibility-v2@e48df452`

Independent source-repair review head: `saq-arbitrary-cardinality-feasibility-v2@c401dae`

CACHE-P authorization: `saq-arbitrary-cardinality-feasibility-v2@ffd0f41`

CACHE-P exact content target: `saq-arbitrary-cardinality-feasibility-v2@56210f8`

CACHE-P independent review record: `saq-arbitrary-cardinality-feasibility-v2@249d5b8`

65. A4 V2: Terminal PAR Failure, Source Repair, And Governance Chain III

CACHE-I authorization/review: `saq-arbitrary-cardinality-feasibility-v2@4f38ca9/@187e363`

CACHE-I exact source/static target: `saq-arbitrary-cardinality-feasibility-v2@c33a2bf`

CACHE-I independent review record: `saq-arbitrary-cardinality-feasibility-v2@5db3025`

PREP authorization target: `saq-arbitrary-cardinality-feasibility-v2@e7f940e`

PREP authorization independent review record: `saq-arbitrary-cardinality-feasibility-v2@16a8201`

Host-rebind erratum target/review: `saq-arbitrary-cardinality-feasibility-v2@5a47fed/@b89dabe`

65. A4 V2: Terminal PAR Failure, Source Repair, And Governance Chain IV

HOST-I-AUTH target: `saq-arbitrary-cardinality-feasibility-v2@212a67b`

HOST-I-AUTH independent review record:
`saq-arbitrary-cardinality-feasibility-v2@71e6bec`

Source-authority erratum target/review:
`saq-arbitrary-cardinality-feasibility-v2@6fe8544/@9fa9528`

HOST-I source-static target/review:
`saq-arbitrary-cardinality-feasibility-v2@4602585/@e8e9c79`

HOST-I-R1 correction-only protocol target/review:
`saq-arbitrary-cardinality-feasibility-v2@ddfef99/@b1a7429`

HOST-I-R1 correction target/review:
`saq-arbitrary-cardinality-feasibility-v2@e17f887/@e10bde7`

65. A4 V2: Terminal PAR Failure, Source Repair, And Governance Chain V

The review preserves the old A4-1S result and changes the next question, not the result. A4-1S timed exact construction together with full canonical research serialization; serialization alone used 18,096 CPU-seconds. That supports neither a solver lower bound nor candidate-only deployment cost.

V2 therefore freezes one primary FOM for the complete four-arm synthetic comparative instrument:

```
T_instrument = C_setup + C_core + C_bundle_io
PASS iff T_instrument <= 34,560,000,000 CPU microseconds
```

65. A4 V2: Terminal PAR Failure, Source Repair, And Governance Chain VI

The cap is only an internal admission boundary for the one frozen synthetic panel. The old `5/2` factor does not transfer: there is no V2 projection to GIST, CIFAR, two datasets, a full vector, an SAQ index build,

or deployment.

Initial source-only inspection found that the parent protocol asked the terminal `P_parity` receipt to include publication work whose terminal values did not yet exist. The additive erratum closes only that self-reference with a finite one-file `PAR_report` after terminal

P. It leaves the three parent blobs, scientific computation, FOM, threshold, retry policy, and status precedence unchanged.

No work disappears from reporting:

65. A4 V2: Terminal PAR Failure, Source Repair, And Governance Chain VII

```
B_build + P_parity + T_instrument
        + E_emit + V_replay + E_archive_body = T_study_metered
PAR_report is a finite one-file PAR authority closure after terminal P
F_trailer is a finite three-file final-study closure after archive-body timin
memory, temporary bytes, bundle bytes, evidence/archive bytes: separately cap
```

Every scientific/parity/control operation and every file validation, discovery, hash, tree-walk, and byte-ledger derivation required to form the seal must finish before terminal P. The post-P closure may only inject captured integers, perform frozen bounded arithmetic, encode one fixed

15-key seal, and atomically publish it. Its exact conservative schema-language maximum is 5,171 bytes including LF; that maximal witness is syntactic, not a semantically admissible resource observation.

65. A4 V2: Terminal PAR Failure, Source Repair, And Governance Chain VIII

The schema freezes 396 atomic construction units, exact-prefix evidence, independent exact-rational scalar/allocation replay, deterministic bit-level block/encoding/packing replay, raw prelaunch evidence, and admissibility-first status precedence. The four-arm bundle is A4-reference-equivalent only; it is not current SAQ's serialized representation or

query estimator.

The corrected protocol authority passed independent timing-closure, schema, and Git-authority review. Initial implementation commit `3a4f7c5` then failed static review and remains non-evidence. Corrected exact source commit `482c401` passed three independent static-only reviews. Its canonical manifest binds 35 exact

sources at source-tree SHA-256 `d5b8374f...`.

65. A4 V2: Terminal PAR Failure, Source Repair, And Governance Chain IX

The user later authorized exactly one frozen PAR invocation at reviewed and pushed base `2e983a0`. It returned exit 3 with frozen status `ARTIFACT_INVALID`: `sys.executable` was `/bin/python`, whose terminal path object was a symlink rejected by

the conductor's no-follow leader-identity read. The failure occurred before host, NumPy, or process-inventory observation and before any B/P worker. No CMake/build child, PCG64 fixture, parity case, receipt, ledger, build manifest, parity summary, artifact index, seal, or final PAR

directory was produced. The empty staging directory and ignored bytecode caches are non-evidence.

The reviewed terminal claim is exactly:

65. A4 V2: Terminal PAR Failure, Source Repair, And Governance Chain X

`ARTIFACT_INVALID`
`REVIEWED_TERMINAL_NO_VALID_PAR_AUTHORITY`
`NO_SCIENTIFIC_DECISION`

This is artifact-validation failure evidence, not `PASS_PARITY`, feasibility, quantization quality, natural-data, ANN, or systems evidence. Static source review did not establish executable-identity compatibility on the pinned host. Tooling correctness remains separate from the database-systems contribution.

The later I-R1 correction changes only CPython leader identity. It opens a stable descriptor for literal `/proc/self/exe`, requires intentionally followed normalized `sys.executable` to name the same nonempty regular inode, performs a bounded full EOF-checked read with before/after

stability, and uses that identity consistently in conductor, runner admission/receipts, and a physically independent positional-read verifier. Generic artifact readers retain `O_NOFOLLOW`.

65. A4 V2: Terminal PAR Failure, Source Repair, And Governance Chain XI

Exact source `e48df452` and canonical 35-source tree `8d8b5d3f8990e5d360e0a617d157a8b934c14d0f37b69f399bcc62c71f98360a` passed three independent static reviews with no finding at LOW or above. No Python module, compiler, build, test, fixture, RNG, native command, or data path was executed. The maximum result

is therefore only:

```
SOURCE_REPAIR_STATIC_REVIEW_PASS
```

65. A4 V2: Terminal PAR Failure, Source Repair, And Governance Chain XII

CACHE-P resolves only the protocol question raised by the five quarantined bytecode files. Its primary-source review records that CPython `-B` suppresses cache writes but does not disable reads of an already present valid cache. The exact target therefore

retains those residuals as unread non-evidence, requires non-destructive sanitized remote isolation, and rejects a direct transition to PAR-R1. It preserves exact source repair `e48df452`, manifest blob `2aa64e50...`, and the 35-source tree `8d8b5d3f8990e5d360e0a617d157a8b934c14d0f37b69f399bcc62c71f98360a`.

The committed CACHE-P claim is exactly:

```
EXACT_TARGET_INDEPENDENT_REVIEW_PASS
GO_SANITIZED_REMOTE_ISOLATION_PROTOCOL_REQUIRED
NO_GO_DIRECT_PAR_R1
```

65. A4 V2: Terminal PAR Failure, Source Repair, And Governance Chain XIII

This is artifact-governance evidence only. CACHE-P imported no module, changed no source or environment, created no clone, prepared no remote tree, and ran no build, fixture, parity, synthetic, base, query, or SAQ work.

CACHE-I implements only the generic cache-governance layer frozen by CACHE-P. Its exact target has 37 filesystem sources (ten Python and 27 native) plus one separately bound inline bootstrap, with source-tree SHA-256 `9568007588c78ddda9fb4c4e20e8773ee2fa1da10a7656f181fef06883de38a2`. The static closure binds four

schema/maximal-instance pairs, cache-aware PAR shapes, the independent cache verifier, and a complete 38-unit import/process/mutation crosswalk. Independent review found 0 issues at LOW or above.

65. A4 V2: Terminal PAR Failure, Source Repair, And Governance Chain XIV

After the source bytes were locked, a separately authorized Python 3.9 `compile()`-only check accepted the six changed/new outer sources and inline bootstrap. No code object was executed and no module was imported. The exact claim is only:

`GENERIC_CACHE_POLICY_SOURCE_STATIC_REVIEW_PASS`

This is artifact-governance source/static-review evidence, not PREP readiness, runtime correctness, parity, feasibility, an SAQ limitation, performance, novelty, or a method.

The later host-rebind erratum `5a47fed/@b89dabe` froze the current `.el9_8.2` leader and a bounded future rebind closure but did not change the source. HOST-I-AUTH target/review `212a67b/@71e6bec` then passed with zero LOW+ findings and only `AUTHORIZATION_EXACT_TARGET_REVIEW_PASS`. It did not edit

65. A4 V2: Terminal PAR Failure, Source Repair, And Governance Chain XV

the five sources or seven derived authority objects, establish HOST_IDENTITY_REBOUND, or run Python, build, PREP, or data. It authorizes no execution. When HOST-I was instructed, exact checking found an unsatisfiable seven-versus-eight protocol-component closure before any source target. Additive

source-authority erratum target/review 6fe8544/@9fa9528 passed three static tracks and reached only HOST_I_SOURCE_AUTHORITY_ERRATUM_REVIEW_PASS. It admits the eighth component but changes no implementation source or derived authority object. The later source-static target 4602585 was independently reviewed at e8e9c79 and failed

with one HIGH authority-identity mismatch: two derived documents cite the wrong governing-review SHA-256. That target remains failed evidence. The correction-only protocol target/review ddfef99/@b1a7429 passed with zero LOW-or-higher findings; exact R1 target/review e17f887/@e10bde7 then passed with zero LOW-or-higher findings

65. A4 V2: Terminal PAR Failure, Source Repair, And Governance Chain XVI

after reproducing the frozen four artifact substitutions, closure, and 13/13 host identities. `HOST_IDENTITY_REBOUND` is established only for the registered 13-object bundle; whole-host identity is not established. Source-history protocol target/review `dcaed57/@e98a3e4` passed. The first authorized implementation stopped before target

on its unsatisfiable ten-path closure, and path-closure erratum target/review `150e5b3/@7749491` passed only for a future eight-path/five-derived-object repair. Exact R1 target/review `e9b5c83/@a1198c4` then reached `SOURCE_HISTORY_EPOCH_CORRECTION_SOURCE_STATIC_REVIEW_PASS`, correcting only the known static verifier rule with zero LOW-or-higher findings. Because the verifier

was not run, cache-verifier/PAR readiness remains unestablished. Fresh R1 PREP authorization target/review `cd1757e/@eb1b893` subsequently passed exact documentation review. Its later unique immediate probe passed, and the unique invocation was consumed but left only durable START and empty captures

65. A4 V2: Terminal PAR Failure, Source Repair, And Governance Chain XVII

before the PREP body. Frozen status is `CRASH_OR_UNKNOWN / NO_SCIENTIFIC_DECISION`. Crash-record protocol target/review `1982865/@8bec546` then passed with zero LOW-or-higher findings; it does not form the record or authorize retry/repair. Audited closure `3577edd` instead stops further artifact-recovery investment at portfolio

level. There is no next authorized research or execution step; reopening requires a new explicit user decision, new question, and fresh cost-justified clean protocol.

Closed downstream stages remain unauthorized and non-transitive:

- **Stage**: `A4-V2-I`
  Scope: source implementation and independent static review only
- **Current authorization**: `COMPLETED_REVIEW_PASS`
- **Stage**: `A4-V2-PAR`
  Scope: frozen build and parity only

65. A4 V2: Terminal PAR Failure, Source Repair, And Governance Chain XVIII

- **Current authorization**: `REVIEWED_TERMINAL_ARTIFACT_INVALID`; no valid PAR authority
- **Stage**: `A4-V2-I-R1`
  Scope: executable-identity source repair and static review only
- **Current authorization**: `SOURCE_REPAIR_STATIC_REVIEW_PASS`
- **Stage**: `A4-V2-CACHE-P`
  Scope: primary-source review and frozen staged-isolation protocol
- **Current authorization**: `EXACT_TARGET_INDEPENDENT_REVIEW_PASS`; governance only
- **Stage**: `A4-V2-CACHE-I`
  Scope: generic source/schema/static closure only
- **Current authorization**: `GENERIC_CACHE_POLICY_SOURCE_STATIC_REVIEW_PASS`; governance only
- **Stage**: historical PREP authorization/review
  Scope: freeze one sanitized remote preparation event

65. A4 V2: Terminal PAR Failure, Source Repair, And Governance Chain XIX

- **Current authorization**: PREP_AUTHORIZATION_EXACT_TARGET_REVIEW_PASS; later nontransferable documentation governance only
- **Stage**: A4-V2-PREP-HOST-P
  Scope: additive current-host rebind protocol and exact review
- **Current authorization**: HOST_IDENTITY_REBIND_PROTOCOL_REVIEW_PASS; host identity is not rebound
- **Stage**: A4-V2-PREP-HOST-I-AUTH
  Scope: freeze only the contract/projection for a possible later source/review edge
- **Current authorization**: AUTHORIZATION_EXACT_TARGET_REVIEW_PASS; documentation governance only; grants no HOST-I authority
- **Stage**: attempted A4-V2-PREP-HOST-I
  Scope: validate the frozen projection before source editing
- **Current authorization**: STOPPED_BEFORE_TARGET; exact seven-versus-eight contradiction

65. A4 V2: Terminal PAR Failure, Source Repair, And Governance Chain XX

- **Stage**: `A4-V2-PREP-HOST-I-ERRATUM`
  Scope: admit the eighth protocol component and freeze exact future accounting
- **Current authorization**: `HOST_I_SOURCE_AUTHORITY_ERRATUM_REVIEW_PASS`; documentation governance only
- **Stage**: `A4-V2-PREP-HOST-I`
  Scope: implement and independently review the coherent five-source/seven-derived-authority rebind
- **Current authorization**: `SOURCE_STATIC_TARGET_REVIEW_FAIL_AUTHORITY_IDENTITY_MISMATCH`; target/review `4602585/@e8e9c79`, one HIGH, retained as immutable failed evidence
- **Stage**: `A4-V2-PREP-HOST-I-R1-P`
  Scope: freeze a correction-only exact five-path target and its direct-child review projection

65. A4 V2: Terminal PAR Failure, Source Repair, And Governance Chain XXI

- **Current authorization**: `HOST_I_R1_CORRECTION_ONLY_REPAIR_PROTOCOL_REVIEW_PASS`; target/review `ddfef99/@b1a7429`, documentation governance only
- **Stage**: `A4-V2-PREP-HOST-I-R1`
  Scope: apply only the frozen authority-identity correction and review it
- **Current authorization**: `HOST_I_R1_CORRECTION_ONLY_REPAIR_REVIEW_PASS`; target/review `e17f887/@e10bde7`; rebound established for registered 13-object bundle only, with no execution
- **Stage**: `A4-V2-SOURCE-HISTORY-P`
  Scope: freeze an exact fail-closed two-epoch correction protocol
- **Current authorization**: `SOURCE_HISTORY_EPOCH_CORRECTION_PROTOCOL_REVIEW_PASS`; target/review `dcaed57/@e98a3e4`, zero LOW+, protocol governance only

65. A4 V2: Terminal PAR Failure, Source Repair, And Governance Chain XXII

- **Stage**: first `A4-V2-SOURCE-HISTORY-I`
  Scope: validate and implement the frozen ten-path projection
- **Current authorization**: `STOPPED_PRE_TARGET_PATH_CLOSURE_UNSATISFIABLE`; runtime schema and maximal witness had no legitimate delta
- **Stage**: `A4-V2-SOURCE-HISTORY-I-R1-P`
  Scope: correct only the future path-closure projection
- **Current authorization**: `SOURCE_HISTORY_I_R1_PATH_CLOSURE_ERRATUM_REVIEW_PASS`; target/review `150e5b3/@7749491`, zero LOW+, protocol governance only
- **Stage**: `A4-V2-SOURCE-HISTORY-I-R1`
  Scope: implement and statically review the corrected exact eight-path/five-derived-object repair

65. A4 V2: Terminal PAR Failure, Source Repair, And Governance Chain XXIII

- **Current authorization**: SOURCE_HISTORY_EPOCH_CORRECTION_SOURCE_STATIC_REVIEW_PASS; target/review e9b5c83/@a1198c4, zero LOW+, no execution or readiness claim
- **Stage**: A4-V2-CACHE-PREP-R1-AUTH
  Scope: freeze the current dormant one-invocation contract and exact review projection
- **Current authorization**: PREP_AUTHORIZATION_EXACT_TARGET_REVIEW_PASS; target/review cd1757e/@eb1b893, zero LOW+, documentation governance only
- **Stage**: A4-V2-CACHE-PREP-ISO-R1
  Scope: run one sanitized preparation after the reserved immediate review-head probe
- **Current authorization**: CRASH_OR_UNKNOWN / START_ONLY / NO_SCIENTIFIC_DECISION; unique probe/invocation consumed before PREP body
- **Stage**: A4-V2-CACHE-PREP-ISO-R1-CRASH-P
  Scope: freeze a truthful future terminal-record topology

65. A4 V2: Terminal PAR Failure, Source Repair, And Governance Chain XXIV

- **Current authorization**: `START_ONLY_CRASH_RECORD_PROTOCOL_REVIEW_PASS`; target/review `1982865/@8bec546`, governance only
- **Stage**: `A4-V2-CACHE-PREP-ISO-R1-CRASH-I`
  Scope: form and independently review the START-only terminal record
- **Current authorization**: `NOT_AUTHORIZED`
- **Stage**: `A4-V2-CACHE-BIND`
  Scope: bind only the independently reviewed sanitized preparation
- **Current authorization**: `NOT_AUTHORIZED`
- **Stage**: `A4-V2-PAR-R1`
  Scope: at most one corrected frozen build/parity event after the full isolation chain
- **Current authorization**: `NOT_AUTHORIZED`; direct transition rejected
- **Stage**: `A4-V2-SRUN`
  Scope: one logical synthetic admission event
- **Current authorization**: `NOT_AUTHORIZED`

65. A4 V2: Terminal PAR Failure, Source Repair, And Governance Chain XXV

- **Stage**: data / SAQ
  Scope: base, query, index, estimator, integration
- **Current authorization**: `NOT_AUTHORIZED`

66. Current Synthesis I

- **Attempt**: 1A full-D PCA replacement
  Theoretical guarantee: L2 isometry for any orthogonal transform
- **Strongest positive evidence**: GIST residual-PCA accurate/full RMSE `-0.60%`/`-0.33%`
  Decision / current boundary: no stable ranking gain; fast harm; CIFAR replication failure
- **Attempt**: 1B lossy `D -> d`
  Theoretical guarantee: exact head and norm terms in favorable oracle
- **Strongest positive evidence**: tail norms reduce RMSE from 0.0445 to 0.00248
  Decision / current boundary: omitted tail IP still worsens ranking versus native SAQ
- **Attempt**: 2 exact scalar DP
  Theoretical guarantee: exact bin-boundary partition SSE using raw bin moments; final midpoint-nearest-centroid raw SSE is evaluated, not reoptimized; outer optimum is conditional on `E[j,b]`

66. Current Synthesis II

- **Strongest positive evidence**: audio B=4 raw MSE `-15.5%`, R@100 `+0.004`
  Decision / current boundary: inner-DP novelty is foreclosed by prior art; no full-scale raw optimum or recall guarantee; cross-regime reversals
- **Attempt**: 3 distance-quality re-evaluation
  Theoretical guarantee: paper-exact metric semantics; no method guarantee
- **Strongest positive evidence**: GIST measured point: higher `1/Ratio`, `1.078x` QPS at the frozen target
  Decision / current boundary: one positive setting; DEEP controls remain negative; metric is prior work
- **Attempt**: 4 arbitrary-cardinality fixed-rate words
  Theoretical guarantee: the arbitrary positive-integer feasible set contains the dyadic set for exact factorized SSE

66. Current Synthesis III

- **Strongest positive evidence**: frozen `(3,5)` witness and A4-1S parity; later positive records are protocol/source/artifact-governance evidence only
  Decision / current boundary: preserve A4-1S `NO_GO_EXACT_SOLVER_COST` and PAR `ARTIFACT_INVALID / NO_SCIENTIFIC_DECISION`; source-history R1 corrected only the static rule; the unique PREP invocation later stopped at START-only `CRASH_OR_UNKNOWN`, and reviewed protocol `1982865/@8bec546` closes only the future record topology; CRASH-I, repair/retry, binding,
- and PAR-R1 remain unauthorized

Cross-attempt lesson:

66. Current Synthesis IV

Optimizing a mathematically valid surrogate is not enough, and changing the evaluation metric is not itself a mechanism.

The missing contribution must connect its optimized quantity to top-k or downstream quality and system work, survive a second baseline or regime, and include all construction, storage, and query overhead.

Attempt 4 was deliberately staged so that a representation theorem or a synthetic witness could not silently become a method claim. The committed and reviewed A4-1S cost stop prevented the next expansion of scope. V2 changes th future measurement contract, its additive erratum closes one reporting self-reference, and its corrected source passed static review. The later PAR failure shows that static review did not guarantee executable-identity compatibility; it is tooling/artifact evidence only and says nothing about

66. Current Synthesis V

instrument feasibility. I-R1 closes the identified source defect under static review, but without corrected execution it still says nothing about parity or instrument feasibility. CACHE-P freezes how a future clean artifact could be isolated, and CACHE-I statically closes only the corresponding generic source and schema layer. Their reviews and the narrow syntax check are governance evidence, not feasibility evidence. The later PREP authorization record also passed exact-target review, but a later static host mismatch retired its

activation path before PREP. The reviewed additive host erratum freezes only a future rebind contract; it does not rebind the host or grant PREP or execution authority. The later HOST-I-AUTH review freezes only the contract and projection for a possible source edge and grants no HOST-I authority; no source/authority rebind, Python, or PREP ran. Its attempted HOST-I edge stopp before edits on a seven-versus-eight component contradiction. The reviewed source-authority erratum fixed that contract admission. The later source-

66. Current Synthesis VI

static target was formed, but its direct-child review failed on one HIGH authority-identity mismatch. Its later correction-only target passed exact review and established rebound only for the registered 13-object bundle. Whole-host and execution readiness remain unestablished. A later reviewed source-history protocol froze a bounded two-epoch repair; after its first ten-path implementation projection failed, an erratum and exact R1 target closed the static eight-path repair under direct-child review. The corrected

verifier was not run, so cache/PAR readiness still cannot be claimed. Fresh R1 PREP authorization target/review `cd1757e/@eb1b893` then passed exact documentation review. The subsequently authorized unique probe/invocation was consumed and stopped at START-only `CRASH_OR_UNKNOWN`; crash-record protocol `1982865/@8bec546` passed, while CRASH-I, repair, retry, and CACHE-BIND remai
unauthorized.

67. Proposed Meeting Discussion I

Decision 1:

`Do the Attempt 1 results justify treating PCA replacement as a closed practic`
`direction, while avoiding a universal PCA-optimality claim?`

Decision 2:

`Should exact-hist DP remain only a stronger offline baseline, given that Wu 1991 / GLM 2017 foreclose the inner algorithmic novelty and White-Singal 2026 explicitly recognizes that baseline, while the local evidence does not establish retrieval or system dominance?`

Decision 3:

`Should future ANN evidence always report both identifier Recall and geometric`
`distance quality, while treating the GIST flip only as measurement evidence?`

67. Proposed Meeting Discussion II

Decision 4:

Do we agree to preserve the terminal A4-1S pipeline-cost failure, while treating the reviewed A4 V2 PAR `ARTIFACT_INVALID` result only as an artifact validation failure, with no inference about arbitrary-cardinality quantization or instrument feasibility; treating I-R1 only as a reviewed source repair rather than artifact readiness; and accepting CACHE-P only at its artifact-governance ceiling, including sanitized remote isolation and the no-go on direct PAR-R1; and treating CACHE-I's static pass as governance

source evidence rather than PREP readiness; and treating the later PREP authorization review as a dormant contract rather than a completed clone or feasibility result; treating HOST-I-AUTH and the source-authority erratum as governance only; and treating the later HOST-I source-static target as a failed authority-DAG artifact whose otherwise passing static checks do not

67. Proposed Meeting Discussion III

establish host rebound by themselves; while treating the reviewed R1 repair PASS as bounded identity governance for the registered 13-object bundle-not

whole-host identity, execution, PREP readiness, or cache-verifier/PAR readiness?

The current authorization boundary is narrow:

A4-V2-PAR was invoked once and terminated before build/parity.
There is no active execution authority and no valid PAR authority.

The clean I-R1 source commit and independent static review are complete.
CACHE-P target/review `56210f8/@249d5b8` froze
`GO_SANITIZED_REMOTE_ISOLATION_PROTOCOL_REQUIRED / NO_GO_DIRECT_PAR_R1`.
CACHE-I target/review `c33a2bf/@5db3025` subsequently reached only

67. Proposed Meeting Discussion IV

`GENERIC_CACHE_POLICY_SOURCE_STATIC_REVIEW_PASS`; the quarantine remains unread and uncleaned. PREP authorization target/review `e7f940e/@16a8201` subsequently reached only `PREP_AUTHORIZATION_EXACT_TARGET_REVIEW_PASS`. Static checking then found the `.el9_8 -> .el9_8.2` leader replacement before PREP or START. Host-rebind erratum target/review `5a47fed/@b89dabe` reached only `HOST_IDENTITY_REBIND_PROTOCOL_REVIEW_PASS`; that protocol stage did not rebind host identity.

The old authority is nontransferable and the old probe is superseded and must never run or be reused. HOST-I-AUTH target/review `212a67b/@71e6bec` reached only `AUTHORIZATION_EXACT_TARGET_REVIEW_PASS`; it performed no five-source or seven-derived-object rebind and ran no Python or PREP. The attempted HOST-I edge then stopped before any source target on an exact seven-versus-eight component contradiction. Source-authority erratum target/review `6fe8544/@9fa9528` reached only

67. Proposed Meeting Discussion V

`HOST_I_SOURCE_AUTHORITY_ERRATUM_REVIEW_PASS`. The subsequently authorized HOST-I target/review `4602585/@e8e9c79` ended at `SOURCE_STATIC_TARGET_REVIEW_FAIL_AUTHORITY_IDENTITY_MISMATCH`: one HIGH, host rebound not established. Correction-only protocol target/review `ddfef99/@b1a7429` then reached `HOST_I_R1_CORRECTION_ONLY_REPAIR_PROTOCOL_REVIEW_PASS`. Exact R1 target/revi `e17f887/@e10bde7` subsequently reached

`HOST_I_R1_CORRECTION_ONLY_REPAIR_REVIEW_PASS`, establishing rebound only for the registered 13-object bundle. Whole-host identity remains unestablished an no Python, build, PREP, or scientific execution ran. A future source-history mismatch then received reviewed protocol target/review `dcaed57/@e98a3e4`. Its first implementation authorization stopped before target on the unsatisfiable ten-path closure; reviewed path-closure erratum `150e5b3/@7749491` froze the exact eight-path/five-derived-object R1 repair.

67. Proposed Meeting Discussion VI

Exact target/review `e9b5c83/@a1198c4` then reached `SOURCE_HISTORY_EPOCH_CORRECTION_SOURCE_STATIC_REVIEW_PASS` with zero LOW+. The verifier was not run, so cache-verifier/PAR readiness is not established. Fresh dormant PREP authorization target/review `cd1757e/@eb1b893` then reached `PREP_AUTHORIZATION_EXACT_TARGET_REVIEW_PASS` with zero LOW+. Its later unique immediate probe and invocation were consumed and left only START plus empty captures before the PREP body, so the result is

`CRASH_OR_UNKNOWN / NO_SCIENTIFIC_DECISION`. Crash-record protocol `1982865/@8bec546` passed independent review. No execution node is currently authorized; stop for a user checkpoint. CRASH-I, source repair/retry, CACHE-BIND, explicit PAR-R1, and SRUN remain separate and unauthorized; base/query inputs, SAQ changes, and the old A4-1 gate remain unauthorized.

Speaker notes:

67. Proposed Meeting Discussion VII

- Attempt 4 completed synthetic correctness validation and a falsification of affordability under the frozen pipeline-cost gate; allocation, block-VQ, and encoding cost phases were not reached after the scalar-prefix early stop.
- A4 V2's scientific PAR work remains unexecuted: the conductor was invoked but stopped at executable-identity admission before build, RNG, or parity. Its erratum repairs reporting closure and its source pass checks artifact machinery only; I-R1 statically repairs the
identified path but has not been executed. The old 5/2 projection is not a V2 cost model.
- Novelty, mechanism, replication, and total overhead should be challenged before experimental scope is expanded.

68. Evidence And Code Map I

Attempt 1, SAQ branch `saq-transform-analysis@3d94840`:

```
docs/saq_transform_phase1_limitation_evidence_2026_07_10.md
docs/saq_transform_phase1b_external_replication_evidence_2026_07_10.md
docs/saq_lossy_projection_lp0_gate_a_evidence_2026_07_11.md
script/prepare_phase1_transform_views.py
src/phase1_transform_diagnostic.cpp
src/lp0_projection_gate_a.cpp
script/evaluate_phase1b_gate.py
```

```
script/evaluate_lp0_gate_a.py
```

Lossy projection evidence is preserved on `saq-lossy-projection-analysis@051ec6a`.

Attempt 2, sibling `vectordb@f51b487`:

68. Evidence And Code Map II

```
src/scalar_quantizer.cpp
src/dimensionwise_quantizer.cpp
src/bit_allocator.cpp
src/eval_dimensionwise_quantizer.cpp
docs/dimensionwise_quantizer_smoke_run_2026_06_26.md
reports/scalar_training_exact_hist_audit_2026_06_30/README.md
reports/scalar_training_exact_hist_audit_2026_06_30/comparison.csv
```

```
reports/scalar_training_exact_hist_audit_2026_06_30/logs/
  deep1M_pca_B4_variable_exact-hist_rank-boundary.stdout
reports/pca_rank_boundary_deep1M_2026_06_27/logs/
  deep1M_pca_B4_variable_lloyd_rank-boundary.stdout
```

Attempt 2 external primary sources:

- White and Singal, Inner Product Aware Quantization, arXiv:2606.00289v1

68. Evidence And Code Map III

- Pinned official code at `e92ed90`
- Grønlund et al., Fast Exact k-Means, k-Medians and Bregman Divergence Clustering in 1D
- Wu, Optimal Quantization by Matrix Searching, Journal of Algorithms, 1991
- Li et al., SAQ: Pushing the Limits of Vector Quantization through Code Adjustment and Dimension Segmentation, arXiv:2509.12086, direct ANN prior for DP segmentation and bit allocation under a space quota

Attempt 3, branch `saq-ratio-metric-analysis@146dc16`:

```
docs/saq_attempt3_ratio_metric_related_work_and_gate_2026_07_13.md
docs/saq_attempt3_ratio_metric_protocol_2026_07_13.md
docs/saq_attempt3_a3_0_a3_1_gist_evidence_2026_07_13.md
docs/saq_attempt3_a3_2_a3_3_decision_2026_07_13.md
docs/saq_attempt3_a3_1b_artifacts_2026_07_13/
```

68. Evidence And Code Map IV

docs/saq_attempt3_a3_2_artifacts_2026_07_13/
script/evaluate_inverse_ratio.py

script/summarize_ratio_frontier.py
src/test_qps.cpp

Attempt 4, branch saq-arbitrary-cardinality-analysis@f1b464b:

docs/saq_attempt4_arbitrary_cardinality_related_work_and_gate_2026_07_13.md
docs/saq_attempt4_arbitrary_cardinality_sources_2026_07_13.json
docs/saq_attempt4_a4_0_offline_protocol_2026_07_13.md
docs/saq_attempt4_a4_0_synthetic_evidence_2026_07_13.md
docs/saq_attempt4_a4_0_artifacts_2026_07_13/synthetic_witness.json
docs/saq_attempt4_a4_1_base_only_feasibility_preregistration_2026_07_13.md
docs/saq_attempt4_a4_1_base_only_input_spec_2026_07_13.json

68. Evidence And Code Map V

```
docs/saq_attempt4_a4_1_base_only_hypotheses_2026_07_13.json
docs/saq_attempt4_a4_1s_synthetic_implementation_protocol_2026_07_13.md
docs/saq_attempt4_a4_1s_artifacts_2026_07_13/parity_artifact_index.json
docs/saq_attempt4_a4_1s_implementation_parity_review_2026_07_13.md
docs/saq_attempt4_a4_1s_artifacts_2026_07_13/synthetic_cost_projection_manife
docs/saq_attempt4_a4_1s_artifacts_2026_07_13/synthetic_cost_projection_summar
docs/saq_attempt4_a4_1s_artifacts_2026_07_13/synthetic_cost_projection_detail

docs/saq_attempt4_a4_1s_artifacts_2026_07_13/cost_projection_artifact_index.j
docs/saq_attempt4_a4_1s_cost_projection_review_2026_07_13.md
```

68. Evidence And Code Map VI

Attempt 4 V2 corrected protocol-authority snapshot `saq-arbitrary-cardinality-feasibility-v2@f13a383`, corrected source snapshot `@482c401`, terminal PAR result/review `@30dfada/@fd5367e`, and exact I-R1 source-repair/review `@e48df452/@c401dae`, followed by CACHE-P exact content target/review record `@56210f8/@249d5b8` and CACHE-I exact source/static target/review `@c33a2bf/@5db3025`,

then the exact PREP authorization target/review `@e7f940e/@16a8201`, and host-rebind erratum target/review `@5a47fed/@b89dabe`, followed by HOST-I-AUTH target/review `@212a67b/@71e6bec`, then source-authority erratum target/review `@6fe8544/@9fa9528`, and finally HOST-I source-static target/review `@4602585/@e8e9c79`, followed by HOST-I-R1 correction-only repair

68. Evidence And Code Map VII

protocol target/review `@ddfef99`/`@b1a7429` and correction target/review `@e17f887`/`@e10bde7`, then source-history epoch-correction protocol target/review `@dcaed57`/`@e98a3e4`, followed by source-history path-closure erratum target/review `@150e5b3`/`@7749491`, and exact source-history R1 target/review `@e9b5c83`/`@a1198c4`, followed by fresh R1 PREP authorization target/review `@cd1757e`/`@eb1b893`,

the consumed START-only invocation, and crash-record protocol target/review `@1982865`/`@8bec546`:

```
docs/saq_a4_v2_primary_source_metadata_2026_07_14.json
docs/saq_a4_v2_cost_evidence_primary_source_review_2026_07_14.md
docs/saq_a4_v2_cost_evidence_go_no_go_memo_2026_07_14.md
docs/saq_a4_v2_synthetic_construction_preregistration_2026_07_14.md
docs/saq_a4_v2_synthetic_construction_contract_2026_07_14.json
docs/saq_a4_v2_artifact_schema_2026_07_14.json
docs/saq_a4_v2_protocol_independent_review_2026_07_14.md
```

68. Evidence And Code Map VIII

docs/saq_a4_v2_par_report_erratum_authorization_2026_07_14.md
docs/saq_a4_v2_par_report_timing_closure_erratum_2026_07_14.md
docs/saq_a4_v2_par_report_timing_closure_erratum_2026_07_14.json
docs/saq_a4_v2_par_report_seal_schema_2026_07_14.json
docs/saq_a4_v2_par_report_seal_maximal_instance_2026_07_14.json
docs/saq_a4_v2_protocol_authority_manifest_2026_07_14.json
docs/saq_a4_v2_par_report_timing_closure_erratum_independent_review_2026_07_1

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68. Evidence And Code Map IX

```
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docs/saq_a4_v2_executable_identity_source_repair_independent_review_2026_07_1
docs/saq_a4_v2_cache_staging_policy_authorization_2026_07_15.md
docs/saq_a4_v2_cache_staging_primary_sources_2026_07_15.json
docs/saq_a4_v2_cache_staging_primary_source_review_2026_07_15.md
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docs/saq_a4_v2_cache_staging_disposition_protocol_independent_review_2026_07_
docs/saq_a4_v2_cache_implementation_authorization_2026_07_15.md
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docs/saq_a4_v2_cache_runtime_schema_2026_07_15.json
docs/saq_a4_v2_cache_prep_binding_schema_2026_07_15.json
```

68. Evidence And Code Map X

docs/saq_a4_v2_isolated_clone_prep_receipt_schema_2026_07_15.json
docs/saq_a4_v2_isolated_clone_prep_token_schema_2026_07_15.json
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docs/saq_a4_v2_prep_host_identity_rebind_erratum_protocol_2026_07_16.md
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68. Evidence And Code Map XI

docs/saq_a4_v2_prep_host_identity_rebind_r1_correction_only_repair_protocol_2
docs/saq_a4_v2_prep_host_identity_rebind_r1_correction_only_repair_contract_2
docs/saq_a4_v2_prep_host_identity_rebind_r1_correction_only_repair_protocol_i
docs/saq_a4_v2_prep_host_identity_rebind_r1_correction_only_repair_independen
docs/saq_a4_v2_source_history_epoch_correction_protocol_2026_07_17.md
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script/a4_v2_cache_policy_verifier.py
docs/saq_a4_v2_source_history_epoch_correction_implementation_independent_rev
docs/saq_a4_v2_prep_start_only_crash_record_protocol_2026_07_18.md
docs/saq_a4_v2_prep_start_only_crash_record_contract_2026_07_18.json

68. Evidence And Code Map XII

`docs/saq_a4_v2_prep_start_only_crash_record_protocol_independent_review_2026_`

A4-0 and A4-1S parity are outcome evidence only at their stated instrument ceilings. A4-1S cost is a reviewed pipeline-cost no-go. A4-1 remains an unexecuted frozen protocol; no natural-data, ANN, or systems result is established. A4 V2 has an

independently reviewed corrected protocol and source plus one reviewed pre-build `ARTIFACT_INVALID` PAR invocation. It produced no valid PAR authority or scientific decision. I-R1 later reached static source- repair review pass only; it did not run. CACHE-P later reached `EXACT_TARGET_INDEPENDENT_REVIEW_PASS`,

requiring sanitized remote isolation and rejecting direct PAR-R1 at an artifact-governance ceiling. It did not read or clean the quarantine. CACHE-I then reached `GENERIC_CACHE_POLICY_SOURCE_STATIC_REVIEW_PASS` for its generic source/schema layer, with a separate syntax-only pass but no import or

68. Evidence And Code Map XIII

code execution. The later PREP authorization record reached `PREP_AUTHORIZATION_EXACT_TARGET_REVIEW_PASS`. A later static host mismatch prevented PREP and START; the old probe is superseded and may never be reused. The host-rebind erratum reached only `HOST_IDENTITY_REBIND_PROTOCOL_REVIEW_PASS`, not host

rebound or PREP readiness. HOST-I-AUTH then reached only `AUTHORIZATION_EXACT_TARGET_REVIEW_PASS`; it changed no source/authority object and ran no Python or PREP. The attempted HOST-I edge stopped before a target on the seven-versus-eight contradiction; its additive source-authority erratum reached

only `HOST_I_SOURCE_AUTHORITY_ERRATUM_REVIEW_PASS` and changed no implementation source or derived authority object. The later HOST-I source-static target/review `4602585/@e8e9c79` failed on one HIGH false governing-review digest, so host rebound remains unestablished and correction was not performed. Correction-only protocol target/review `ddfef99/@b1a7429` then

68. Evidence And Code Map XIV

passed with zero LOW-or-higher findings. Exact R1 target/review `e17f887/@e10bde7` subsequently passed and established rebound only for the registered 13-object bundle; whole-host identity remains unestablished and no execution ran. A separate `SOURCE_HISTORY_MISMATCH` source repair was still required before cache-verifier

or PAR readiness; its protocol target/review `dcaed57/@e98a3e4` passed, the first implementation authorization stopped before target on an unsatisfiable ten-path projection, and path-closure erratum target/review `150e5b3/@7749491` passed only for a corrected future eight-path repair. Exact R1 target/review `e9b5c83/@a1198c4` then

corrected only that static rule and reached `SOURCE_HISTORY_EPOCH_CORRECTION_SOURCE_STATIC_REVIEW_PASS` with zero LOW+. The verifier was not run, so readiness remains unestablished. Fresh dormant PREP authorization target/review `cd1757e/@eb1b893` then passed exact documentation review with zero LOW+. Its later unique immediate

68. Evidence And Code Map XV

probe passed, but the consumed invocation left only durable START plus empty captures before the PREP body. Frozen status is `CRASH_OR_UNKNOWN / NO_SCIENTIFIC_DECISION`. Crash-record protocol target/review `1982865/@8bec546` then passed with zero LOW+, freezing only the future record topology. CRASH-I,

repair/retry, CACHE-BIND, PAR-R1, A4-V2-SRUN, and all data access remain unauthorized until their own reviewed and explicit authorities exist.

Current deck:

`docs/saq_next_meeting_attempts_1_4_slides_2026_07_13.md`

69. One-Slide Takeaway I

Attempt 1:
PCA replacement found a small GIST estimator signal that failed CIFAR replication. The favorable lossy D->d oracle still ranked worse than native SAQ because a norm-only tail cannot recover tail inner products.

Attempt 2:
The paper and pinned public code do not present an integrated shared-codebook

heterogeneous-rate, outer-allocation pipeline; that absence does not establis
novelty. Exact 1D scalar DP is old prior art. Separately, the pinned code exposes a full-raw-data exact solver primitive and an adjacent shared-block helper, neither with a public integrated experiment path. Local raw-SSE gains are real, but Recall can improve or regress. Keep this as offline baseline an
objective-mismatch evidence, not a method.

69. One-Slide Takeaway II

```
Attempt 3:
Paper-exact 1/Ratio changes one GIST matched-quality conclusion: a measured
fac-error point gives higher geometric quality and 1.078x QPS. DEEP B=4/B=5
remain negative, 39.6% of paired GIST queries worsen, and the metric is prior
work. The result is measurement evidence, not a method.

Attempt 4:

Fixed-rate arbitrary cardinalities strictly beat the dyadic scalar-product
restriction on the frozen (3,5) witness, but unrestricted block VQ matches
them and the primitives are known. A4-0 validates only the instrument. A4-1S
exact implementation parity passed, but the frozen construction/evidence
pipeline projected to 24.170246892361 CPU-hours and failed its 24-hour gate.
That terminal result remains unchanged. A4 V2 now freezes a direct internal
comparative-instrument FOM and a complete evidence ledger. Its reviewed
```

69. One-Slide Takeaway III

erratum adds only a finite one-file terminal-P reporting closure; the old 5/2 projection does not transfer. Corrected V2 source passed independent static review. The later authorized PAR conductor ran only far enough to reject the `/bin/python` symlink at its no-follow identity boundary; no build, RNG, fixture, parity case, or final artifact was produced. I-R1 subsequently bound leader identity to stable `/proc/self/exe` bytes plus same-inode `sys.executable` validation and passed exact-commit static review. It was not

executed and therefore establishes neither parity nor artifact readiness. CACHE-P subsequently reviewed CPython cache semantics and the frozen source without reading the quarantine. Exact target `56210f8`, independently reviewed at `249d5b8`, freezes sanitized remote isolation and rejects direct PAR-R1. That is artifact governance only: it creates no clone, preparation, parity, o scientific evidence. CACHE-I target `c33a2bf`, independently reviewed at `5db3025`, then closed 37 filesystem sources and one separate inline bootstra

69. One-Slide Takeaway IV

under the generic cache policy. Its separately authorized syntax-only check executed no code object. The result is only `GENERIC_CACHE_POLICY_SOURCE_STATIC_REVIEW_PASS`, not PREP readiness or feasibility evidence. PREP authorization target `e7f940e`, independently reviewed at `16a8201`, subsequently reached only `PREP_AUTHORIZATION_EXACT_TARGET_REVIEW_PASS`. Static prelaunch checking then found the `.el9_8 -> .el9_8.2` leader replacement before PREP or START. The

host-rebind erratum target `5a47fed`, independently reviewed at `b89dabe`, reached only `HOST_IDENTITY_REBIND_PROTOCOL_REVIEW_PASS`. It created no clone token, or receipt, did not rebind the host, and superseded the old probe; HOST-I-AUTH target `212a67b`, independently reviewed at `71e6bec`, then reached only `AUTHORIZATION_EXACT_TARGET_REVIEW_PASS`. That is documentation governance only: no five-source/seven-derived-object rebind, Python, build, o PREP occurred. The attempted HOST-I source edge stopped before a target on an

69. One-Slide Takeaway V

exact seven-versus-eight component contradiction. Source-authority erratum target `6fe8544`, independently reviewed at `9fa9528`, reached only `HOST_I_SOURCE_AUTHORITY_ERRATUM_REVIEW_PASS`; it changes no implementation source or derived authority object. The later source-static target `4602585`, independently reviewed at `e8e9c79`, failed with one HIGH because two authority documents cite the wrong governing-review SHA-256. Host rebound is not established. Correction-only protocol target `ddfef99`, independently

reviewed at `b1a7429`, froze the exact five-path R1 edge. Correction target `e17f887`, independently reviewed at `e10bde7`, then passed with zero LOW-or-higher findings across the correction, closure, and 13/13 registered host objects. This establishes only registered-bundle rebound, not whole-host or execution readiness. Source-history protocol target `dcaed57`, independently reviewed at `e98a3e4`, subsequently passed and froze a strict two-epoch futur repair. The first implementation authorization stopped before target on an

69. One-Slide Takeaway VI

unsatisfiable ten-path closure. Path-closure erratum target `150e5b3`, independently reviewed at `7749491`, then passed only for a corrected future eight-path/five-derived-object repair. Exact R1 target `e9b5c83`, independent reviewed at `a1198c4`, then implemented only that correction and reached `SOURCE_HISTORY_EPOCH_CORRECTION_SOURCE_STATIC_REVIEW_PASS`. It ran no verifier, build, PREP, or data, so cache-verifier/PAR readiness is still unestablished. Fresh dormant PREP authorization target `cd1757e`, independent

reviewed at `eb1b893`, then reached `PREP_AUTHORIZATION_EXACT_TARGET_REVIEW_PASS` with zero LOW-or-higher findings. Its later unique probe/invocation was consumed and stopped at START-only `CRASH_OR_UNKNOWN` before the PREP body. Crash-record protocol target/review `1982865/@8bec546` then passed, defining only a future terminal record topology and no scientific or retry/repair authority.

69. One-Slide Takeaway VII

Project decision:
Keep Attempts 1--3 as rigorous negative or partial evidence. Do not rescue them with post-hoc sweeps. Preserve A4-1S as a preregistered synthetic cost stop. Treat A4 V2 only as reviewed protocol plus a source-static review pass, followed by a reviewed terminal `ARTIFACT_INVALID` and a later reviewed stati source repair, not feasibility evidence. CACHE-P additionally provides a reviewed isolation protocol, with

`GO_SANITIZED_REMOTE_ISOLATION_PROTOCOL_REQUIRED / NO_GO_DIRECT_PAR_R1`, but the later CACHE-I result remains static artifact governance only. Treat the subsequent PREP authorization and host-rebind erratum reviews as documentatic governance, not PREP readiness or execution. Treat HOST-I-AUTH the same way: it freezes only the contract/projection for a possible future source edge and grants no HOST-I authority. Treat the source-authority erratum as a correctic of that projection only, not source implementation, host rebound, or

69. One-Slide Takeaway VIII

readiness. Treat the later HOST-I target as a failed authority-DAG target, nc
host rebound. Its later correction-only protocol passed direct-child review,
and actual R1 subsequently passed direct-child review, establishing only the
registered 13-object bundle identity. Do not promote that result to whole-hos
PREP, cache/PAR, performance, or scientific readiness. Treat the reviewed
source-history protocol and path-closure erratum as correction governance;
the first implementation attempt stopped before target, while the later exact

R1 correction passed direct-child static review only. Do not promote that
source result to executed verifier, PREP, cache/PAR, performance, or
scientific readiness. Treat fresh R1 PREP authorization target/review
`cd1757e/@eb1b893` as the reviewed authority for the now-consumed unique
probe/invocation. That invocation produced only START-only
`CRASH_OR_UNKNOWN`, not PREP readiness or science. Treat reviewed protocol
`1982865/@8bec546` as record governance only. CRASH-I, source repair/retry,

69. One-Slide Takeaway IX

`CACHE-BIND, separately explicit PAR-R1, and SRUN remain a strictly ordered unauthorized chain; base, query, and SAQ gates remain closed.`